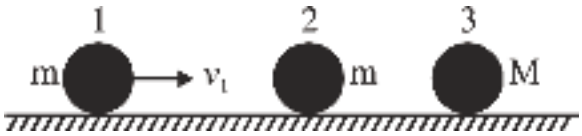
ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

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04-10-2026
MD

PHYSICS

1) Two balls marked 1 and 2 of the same mass m and a third ball marked 3 of mass M are arranged over a smooth horizontal surface as shown in figure. Ball 1 moves with a velocity v_1 towards ball 2 and 3. All collisions are assumed to be elastic. If M < m, the number of collisions between the balls will be :-



(1) One

(2) Two

(3) Three

(4) Four

2) A shell of mass 200 g is ejected from a gun of mass 4 kg by an explosion that generates 1.05 kJ of energy. The initial velocity of the shell is

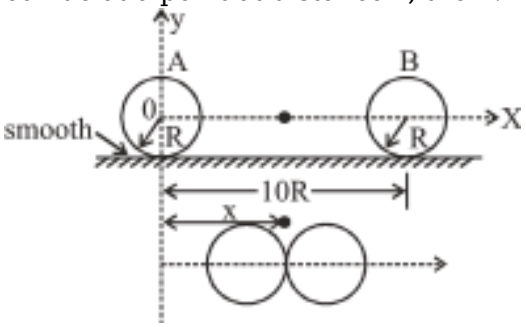
(1) 40 m/s

(2) 120 m/s

(3) 100 m/s

(4) 80 m/s

3) Two uniform non conducting balls A and B have identical size having radius R but made of different density material (density of A = 2 × density of B). The ball A is +vely charged and ball B is –vely charged. The balls are released on the horizontal smooth surface at the separation 10R as shown in figure. Because of mutual attraction the balls start moving towards each other. They will collide at a point at distance x, then :-



(1) $x = \frac{10R}{3}$

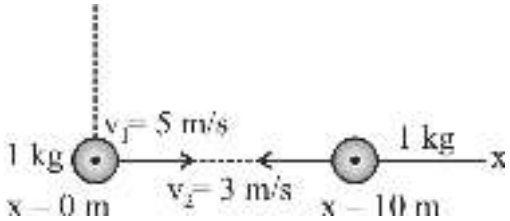
(2) $x = \frac{11R}{3}$

(3) $x = 5R$

(4) $x = \frac{7R}{3}$

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4) At t = 0, the positions and velocities of two particles are as shown in the figure. They are kept on a smooth surface and being mutually attracted by gravitational force. Find the position of centre of mass at t = 2s :-



(1) X = 5 m

(2) X = 7 m

(3) X = 3 m

(4) X = 2 m

5) A body of mass 2kg moves horizontally with some speed towards a stationary body for elastic collision, 2 kg body rebounds with $\frac{1}{3}$rd of its original speed then find mass of second body ?

(1) 1 kg

(2) 4 kg

(3) 3 kg

(4) None

6) A circular disc of radius R is removed from a bigger circular disc of radius 2R such that the circumferences of the discs concide. The centre of mass of the remaining part is at a distance of αR from the centre of the bigger disc. The value of α is :-

(1) $\frac{1}{3}$

(2) $\frac{1}{2}$

(3) $\frac{1}{6}$

(4) $\frac{1}{4}$

7) A body of mass 4 kg moving with velocity 12 m/s collides with another body of mass 6 kg at rest. If both bodies stick together after collision, then the loss of kinetic energy of system in collision is :-

(1) zero

(2) 288 J

(3) 172.8 J

(4) 144 J

8) Two identical balls A and B having velocities of 0.5 m/s and –0.3 m/s respectively collide elastically in one dimension. The velocities of B and A after the collision respectively will be :

(1) 0.3 m/s and 0.5 m/s

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(2) –0.5 m/s and 0.3 m/s

(3) 0.5 m/s and –0.3 m/s

(4) –0.3 m/s and 0.5 m/s

9) A man of 50 kg mass is standing in a gravity free space at a height of 10 m above the floor. He throws a stone of 0.5 kg mass downwards with a speed 2 m/s. When the stone reaches the floor, the distance of the man above the floor will be :-

(1) 20 m

(2) 9.9 m

(3) 10.1 m

(4) 10 m

10) The mass per unit length of a non-uniform rod of length L is given by $\mu = \lambda x^2$, where λ is a constant and x is distance from one end of the rod. The distance of the centre of mass of rod from this end is : -

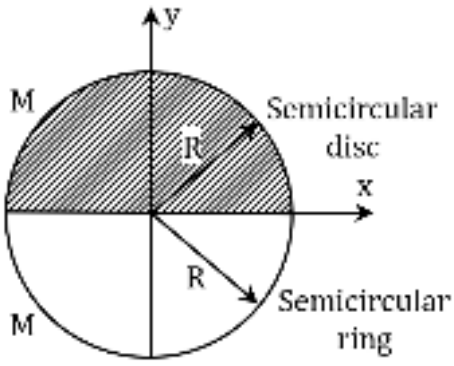
(1) $\frac{L}{2}$

(2) $\frac{L}{4}$

(3) $\frac{3L}{4}$

(4) $\frac{L}{3}$

11) A semicircular disc & semicircular ring of radius R = 9πm having same mass M are placed as shown then coordinate of centre of mass of system will be : -



(1) (0, 15)

(2) (0, 3)

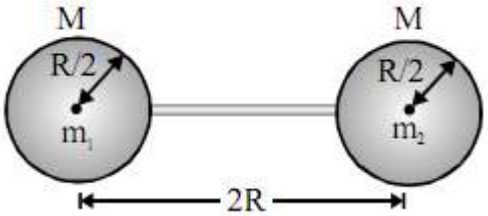
(3) (0, –3)

(4) (0, –6)

12) Two spheres each of mass M and radius R/2 are connected with a massless rod of length 2R as shown in the figure. What will be the moment of inertia of the system about an axis passing through

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the centre of one of the spheres and perpendicular to the rod :-



(1) $\frac{21}{5}MR^2$

(2) $\frac{2}{5}MR^2$

(3) $\frac{5}{2}MR^2$

(4) $\frac{5}{21}MR^2$

13) A wheel of mass 10 kg has a moment of inertia of 160 kg-m^2 about its own axis, the radius of gyration will be : -

(1) 10 m

(2) 8 m

(3) 6 m

(4) 4 m

14) A door 1.6m wide requires a force of 1N to be applied at the free end to open or close it. The force that is required at a point 0.4m distant from the hinges for opening or closing the door is : -

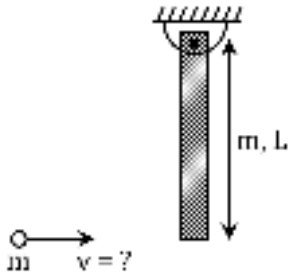
(1) 1.2N

(2) 3.6N

(3) 2.4N

(4) 4N

15) A ball of mass m is moving with velocity v_0 is strike at force end of a rod of mass m and length L hinged as shown in figure and stick to it. What will be the velocity of ball so that rod plus ball will be in horizontal position at their maximum displacement?



(1) $\sqrt{3gL}$

(2) $\sqrt{2gL}$

(3) $2\sqrt{gL}$

(4) None

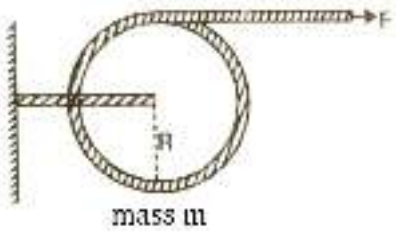
16) If a solid sphere is rolling, then the ratio of its rotational kinetic energy to the total kinetic

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energy is : -

(1) 1 : 2
(2) 2 : 5
(3) 2 : 7
(4) 5 : 7

17) A disc of mass m and radius R is hinged at its centre. A light string is wrapped on its rim and the free end of the string is being pulled by a force F. Assuming that the string does not slip over the disc, the acceleration of free end of the string is : -



(1) $\frac{F}{m}$
(2) $\frac{F}{3m}$
(3) $\frac{2F}{m}$
(4) $\frac{F}{2m}$

18) A person sitting firmly over a rotating stool has his arms stretched if he folds his arms, his angular momentum about the axis of rotation : -

(1) increases
(2) decreases
(3) remains unchanged
(4) doubles

19) A ring of mass 10 kg and diameter 0.4m is rotated about its axis. If it makes 2100 revolutions per minute, then its angular momentum will be : -

(1) 44 kg × m^2/s
(2) 88 kg × m^2/s
(3) 4.4 kg × m^2/s
(4) 0.4 kg × m^2/s

20) If a solid sphere of mass 1 kg and radius 0.1 m rolls without slipping at a uniform velocity of 1 m/s along a straight line on a horizontal floor, the kinetic energy is :

(1) $\frac{7}{5}$J

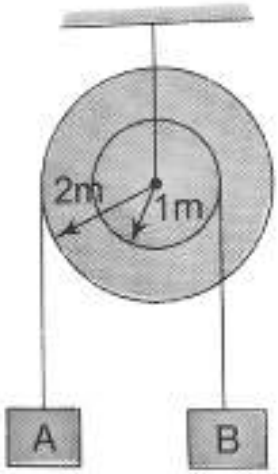
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(2) $\frac{2}{5}$J

(3) $\frac{7}{10}$J

(4) 1J

21) In the pulley system shown, if radii of the bigger and smaller pulley are 2m and 1m respectively and the acceleration of block A is 5 m/s^2 in the downward direction, then the acceleration of block B will be :



(1) 0 m/s^2

(2) 5 m/s^2

(3) 10 m/s^2

(4) $\frac{5}{2}$ m/s^2

22) A grindstone increases in angular speed from 4.00 rad/s to 12.00 rad/s in 4.00 s. Through what angle does it turn during that time interval if the angular acceleration is constant ?

(1) 8.00 rad

(2) 12.00 rad

(3) 16.00 rad

(4) 32.00 rad

23) A metre stick is balanced on a knife edge at its centre. When two coins, each of mass 5g are put one on of the other at the 12 cm mark, the stick is found to balanced at 45 cm. The mass of the metre stick is

(1) 56 g

(2) 66 g

(3) 76 g

(4) 36 g

24) For a particle of a mass 100 gm, position and velocity at any instant are given as $10\hat{i} + 6\hat{j}$ cm and $\vec{v} = 5\hat{i}$ cm/s. Calculate the angular momentum about the point (1, 1) cm.

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(1) $25 \times 10^{-5}\left(-\hat{k}\right)$ kgm^{2}s^{-1}

(2) $30 \times 10^{-5}\left(-\hat{k}\right)$ kgm^{2}s^{-1}

(3) $3 \times 10^{-5}\left(-\hat{k}\right)$ kgm^{2}s^{-1}

(4) $40 \times 10^{-4}\left(-\hat{k}\right)$ kgm^{2}s^{-1}

25) A loop and a disc have same mass and roll without slipping with the same linear velocity v. If the total kinetic energy of the loop is 8 J, the kinetic energy of the disc must be

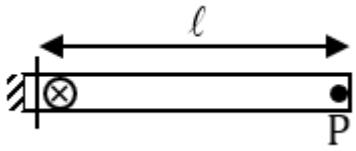
(1) 8 J

(2) 16 J

(3) 6 J

(4) 4 J

26) A uniform rod smoothly pivoted at one of its ends is released from rest. If it swings in vertical plane, the maximum speed of the end P of the rod is :



(1) $2\sqrt{3g\ell}$

(2) $\sqrt{3g\ell}$

(3) $2\sqrt{2g\ell}$

(4) $\sqrt{g\ell}$

27) **Assertion :** A disc can not perform pure rolling motion on a smooth inclined plane.
Reason : In pure rolling the point of contact is at rest with respect to contact surface.

(1) If both assertion and reason are true and reason is the correct explanation of assertion.

(2) If both assertion and reason are true but reason is not the correct explanation of assertion.

(3) If assertion is true but reason is false.

(4) If both assertion and reason are false.

28) A wheel having moment of inertia 2 kg–m^2 about its vertical axis, rotates at the rate of 60 rpm about the axis. The torque which can stop the wheel's rotation in one minute would be : -

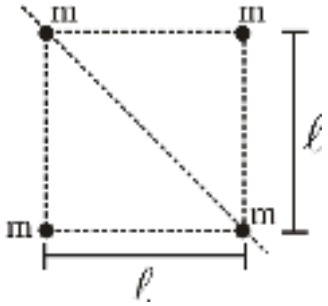
(1) π/12 N–m

(2) π/15 N–m

(3) π/18 N–m

(4) 2π/15 N–m

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29) Find moment of inertia of system about diagonal of square :-

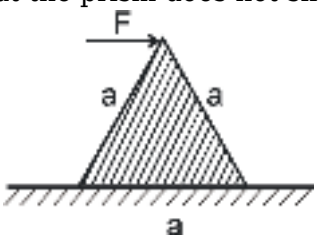
(1) $2m\square^2$

(2) $m\square^2$

(3) $\frac{m\ell^2}{2}$

(4) $\frac{m\ell^2}{\sqrt{2}}$

30) An equilateral uniform prism of mass m rests on a rough horizontal surface with coefficient of friction μ. A horizontal force F is applied on the prism as shown in the figure. If the coefficient of friction is sufficiently high so that the prism does not slide before toppling, then the minimum force required to topple the prism is :



(1) $\frac{mg}{\sqrt{3}}$

(2) $\frac{mg}{4}$

(3) $\frac{\mu mg}{\sqrt{3}}$

(4) $\frac{\mu mg}{4}$

31) The moment of inertia of a uniform circular disc of radius R and mass M about an axis touching the disc at its diameter and normal to the disc is

(1) MR^2

(2) $\frac{2}{5}MR^2$

(3) $\frac{3}{2}MR^2$

(4) $\frac{1}{2}MR^2$

32) A rigid body is rotating such that angular displacement at any instant t is given by $\theta = 2t^2 + 3t + 4$

	Column I		Column II

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(P)	Angular velocity at t = 2 units	(1)	4 units
(Q)	Angular acceleration at t = 3 units	(2)	0 units
(R)	Average angular velocity between t = 0 and t = 2 units	(3)	7 units
(S)	Average angular acceleration between t = 0 and t = 2 units	(4)	11 units

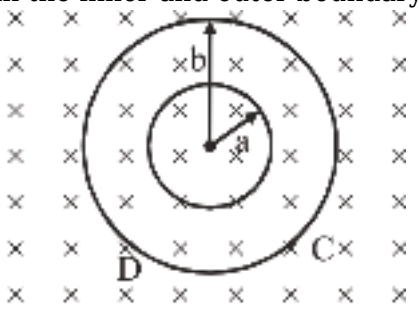
(1) P→4;Q→1;R→3;S→1

(2) P→3;Q→1;R→4;S→1

(3) P→1;Q→1;R→3;S→4

(4) P→4;Q→3;R→1;S→1

33) Plane figures made of thin wires of resistance R = 50 milli ohm/metre are located in a uniform magnetic field perpendicular into the plane of the figures and which decrease at the rate dB/dt = 0.1 T/s. Then currents in the inner and outer boundary are. (The inner radius a = 10 cm and outer radius b = 20 cm)



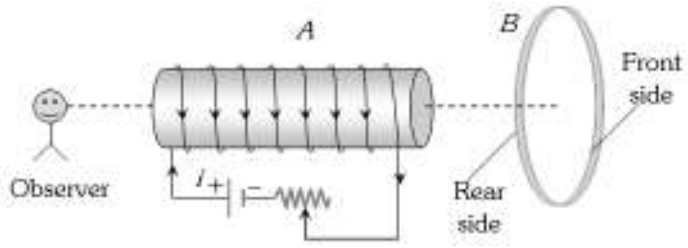
(1) 10^{-1} A (Clockwise), 2×10^{-1} A (Clockwise)

(2) 10^{-1} A (Anticlockwise), 2×10^{-1} A (Clockwise)

(3) 2×10^{-1} A (Clockwise), 10^{-1} A (Anticlockwise)

(4) None of these

34) An aluminium ring *B* faces an electromagnet *A*. The current *I* through *A* can be altered



(1) Whether *I* increases or decreases, *B* will not experience any force

(2) If *I* decrease, *A* will repel *B*

(3) If *I* increases, *A* will attract *B*

(4) If *I* increases, *A* will repel *B*

35) The flux linked with a circuit is given by $\phi = t^3 + 3t - 7$. The graph between time (x-axis) and induced emf (y-axis) will be

(1) Straight line through the origin

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(2) Straight line with positive intercept

(3) Straight line with negative intercept

(4) Parabola not through the origin

36) A coil of N = 100 turns carries a current I = 5 A and creates a magnetic flux $\phi = 10^{-5}$ Tm^{-2} per turn. The value of its inductance L will be

(1) 0.05 mH

(2) 0.10 mH

(3) 0.15 mH

(4) 0.20 mH

37) A square of side □ meters lies in the x-y plane in a region, where the magnetic field is given by $\vec{B} = \left(a\hat{i} + b\hat{j} + c\hat{k}\right)$ T, where a,b,c are constant. The magnitude of flux in Wb passing through the square is.

(1) □^{2}b

(2) □^{2}a

(3) □a^2

(4) □^{2}c

38) A solenoid having 500 turns and length 2 m has radius of 2 cm, then self inductance of solenoid ?

(1) 4×10^{-4} H

(2) 2×10^{-4} H

(3) 8×10^{-4} H

(4) 16×10^{-4} H

39) A copper disc of radius 0.1 m rotates about its centre with 10 revolutions per second in a uniform magnetic field (along the axis of disc) of 0.1 Tesla. The emf induced across the radius of the disc is :-

(1) $\frac{\pi}{10}$V

(2) $\frac{2\pi}{10}$V

(3) 10 πmV

(4) 20 πmV

40) An equilaterial triangular loop having a resistance R and length of each side '□' is placed in a magnetic field which is varying at $\frac{dB}{dt} = 1$ T/s. The induced current in the loop will be :-

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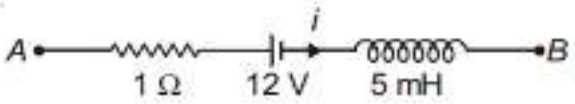
(1) $\frac{\sqrt{3}}{4}\frac{\ell^2}{R}$

(2) $\frac{4}{\sqrt{3}}\frac{\ell^2}{R}$

(3) $\frac{\sqrt{3}}{4}\frac{R}{\ell^2}$

(4) $\frac{4}{\sqrt{3}}\frac{R}{\ell^2}$

41) The network shown in figure is a part of a complete circuit. If at a certain instant, the current i is 4 A and is increasing at a rate of 10^3 A/s. Then $V_B - V_A$ will be



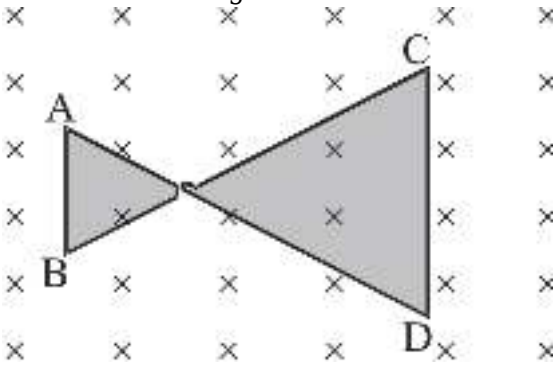
(1) –11 V

(2) 11 V

(3) –21 V

(4) 21 V

42) A conducting wire frame is placed in a magnetic field, which is directed into the paper. The magnetic field is increasing at a constant rate. The directions of induced currents in wires AB and CD are :-



(1) A to B and C to D

(2) B to A and C to D

(3) A to B and D to C

(4) B to A and D to C

43) In LR circuit the current increases to one fourth of its maximum value in 4sec, then the time constant of the circuit is -

(1) $2\log_e 2$

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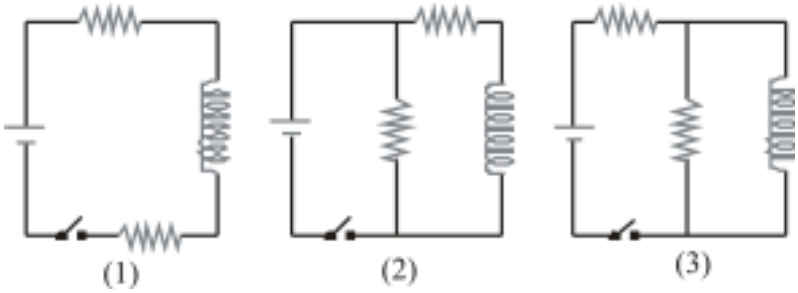
(2) $\frac{4}{\log_e 2}$

(3) $\frac{4}{\log_e\left(\frac{4}{3}\right)}$

(4) $\frac{\log_e 2}{2}$

44)

The figure shows three circuits with identical batteries, inductors, and resistors. Rank the circuits according to the current through the battery
(i) just after the switch is closed and
(ii) a long time later, greatest first



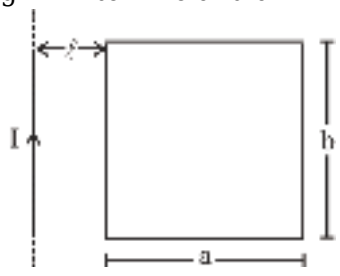
(1) (i) $i_2 > i_3 > i_1\ (i_1 = 0)$ (ii) $i_2 > i_3 > i_1$

(2) (i) $i_2 < i_3 < i_1 (i_1 \neq 0)$ (ii) $i_2 > i_3 > i_1$

(3) (i) $i_2 = i_3 = i_1 (i_1 = 0)$ (ii) $i_2 < i_3 < i_1$

(4) (i) $i_2 = i_3 > i_1 (i_1 \neq 0)$ (ii) $i_2 > i_3 > i_1$

45) The coefficient of mutual induction (M) between a current carrying infinite wire and a rectangular loop of side 'a' and 'b' kept at a distance '' from wire is :-



(1) $\frac{\mu_0 I}{2\pi}\ln\left(1+\frac{a}{\ell}\right)$

(2) $\frac{\mu_0 b^2}{2\pi a}\ln\left(1+\frac{a}{\ell}\right)$

(3) $\frac{\mu_0 a^2}{2\pi b}\ln\left(1+\frac{a}{\ell}\right)$

(4) $\frac{\mu_0 b}{2\pi}\ln\left(1+\frac{a}{\ell}\right)$

CHEMISTRY

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1) Which one is most reactive towards S_N1 reaction?

(1) $C_6H_5CH(C_6H_5)Br$
(2) $C_6H_5CH(CH_3)Br$
(3) $C_6H_5C(CH_3)(C_6H_5)Br$
(4) $C_6H_5CH_2Br$

2)

The above reaction is called -
$C_2H_5Cl + AgF \rightarrow C_2H_5F + AgCl$

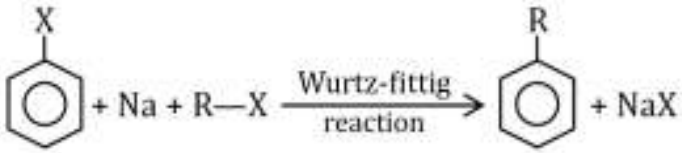
(1) Williamson synthesis
(2) Swarts reaction
(3) Wurtz fittig reaction
(4) Wurtz reaction

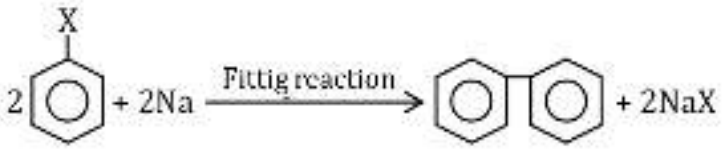
3) The reactivity of alkyl halides in E_2 elimination reactions follows the order

(1) R – I < R – Br < R – Cl < R – F
(2) R – F < R – Cl < R – Br < R – I
(3) R – I > R – Cl > R – Br < R – F
(4) R – I < R – Br < R – F < R – Cl

4) Which of the following is correct

(1) $2RX + 2Na \xrightarrow{\text{Wurtz reaction}} R—R + 2NaX$

(2) 

(3) 

(4) All of these

5) Which of the following reaction is correct :

(1) $CH_3CH_2—Br + KCN \rightarrow CH_3CH_2—NC$
(2) $CH_3CH_2—Br + AgCN \rightarrow CH_3CH_2—CN$
(3) $CH_3CH_2—Br + KNO_2 \rightarrow CH_3—CH_2—N{=}O$ (O)
(4) $CH_3CH_2—Br + AgNO_2 \rightarrow CH_3—CH_2—N{=}O$ (O)

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6) CH_3–CH_2–CH_2–OH $\xrightarrow[\text{reagent}]{\text{Jone's}}$ (P) $\xrightarrow{PCl_5}$ (Q) $\xrightarrow{H_2+Pd+BaSO_4}$ (R)
Product P, Q and R respectively.

(1) CH_3CH_2COOH, CH_3CH_2COCl, $CH_3CH_2CH_2Cl$

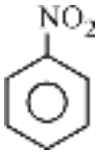
(2) CH_3CH_2COOH, CH_3CH_2CHO, CH_3CH_2COCl

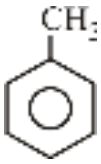
(3) CH_3CH_2COOH, CH_3CH_2COCl, CH_3CH_2CHO

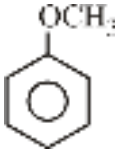
(4) CH_3COOH, CH_3COCl, CH_3CHO

7)

Most Reactive species towards Electrophilic Substitution Reaction is:-

(1) 

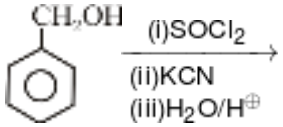
(2) 

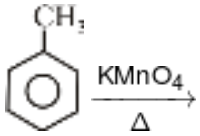
(3) 

(4) 

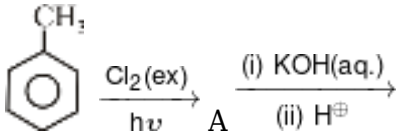
8)

Dr. Sandy Gilbert asked his students to prepare Benzoic acid by different methods, find out number of correct methods :

(a) Kushal's method : 

(b) Ayushi's method : 

(c) Jugan Jyot's method : Ph Mg Br $\xrightarrow[(ii)H_2O]{(i)CO_2}$

(d) Anu Priya's method : 

(1) One

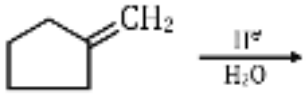
(2) Two

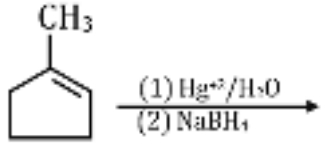
Watermarkly

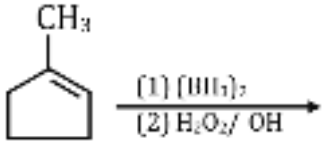
(3) Three

(4) Four

9) In which reaction 3° Alcohol is the major product ?

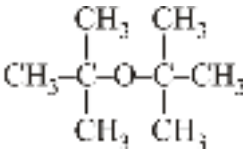
(1) 

(2) 

(3) 

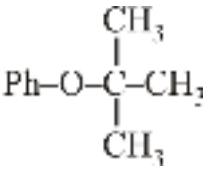
(4) Both (1) and (2)

10) Which of the following ether can be prepared by Williamson synthesis reaction ?

(1) 

(2) Ph–O–Ph

(3) Ph–O–CH_3

(4) 

11) Secondary alcohol on heating with copper at 300°C give.

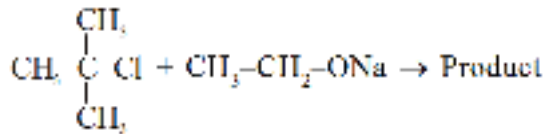
(1) Alkene

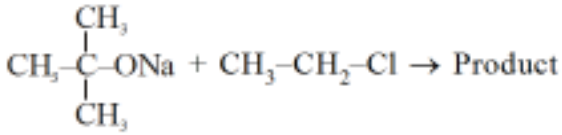
(2) Aldehyde

(3) Ketone

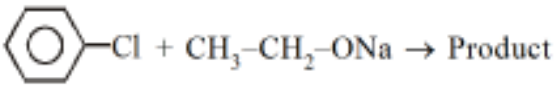
(4) tert. alcohol

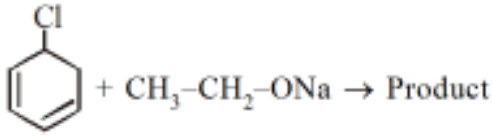
12) Which of the following reaction is most suitable for preparation of ether :

(1) 

(2) 

Watermarkly

(3) 

(4) 

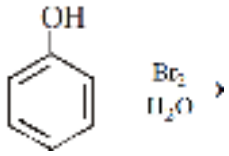
13) Phenol, when it first reacts with concentrated sulphuric acid and then with concentrated nitric acid, gives

(1) 2, 4, 6-trinitrobenzene

(2) picric acid

(3) p-nitrophenol

(4) nitrobenzene

14)  Product formed is :

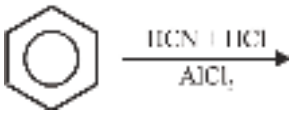
(1) 2,4,6-Tribromoaniline

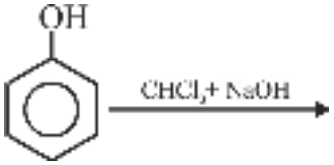
(2) 2-Bromophenol and 4-Bromophenol

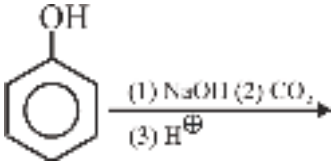
(3) Picric acid

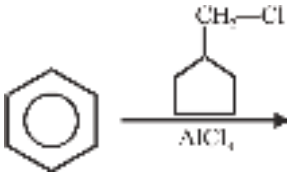
(4) 2,4,6-Tribromophenol

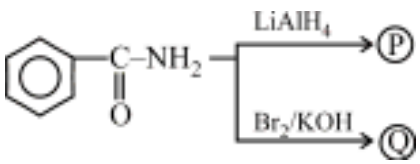
15) Which of the following is Reimer-Tiemann reaction :-

(1) 

(2) 

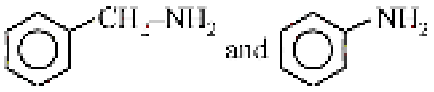
(3) 

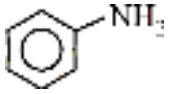
(4) 

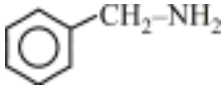
16) 

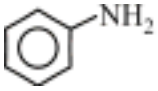
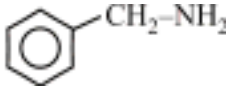
Watermarkly

Product P and Q is :

(1)  and

(2) Both 

(3) Both 

(4)  and 

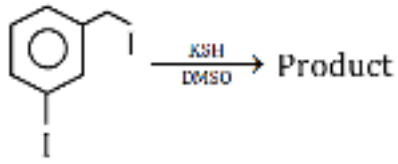
17) Correct order of nucleophilicity ?

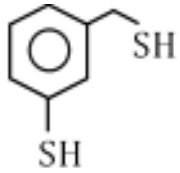
(1) $CH_3S^- > CH_3O^- > CH_3\text{–}COO^- > F^-$

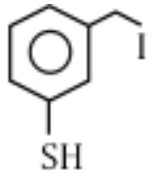
(2) $CH_3\text{–}COO^- > CH_3S^- > CH_3O^- > F^-$

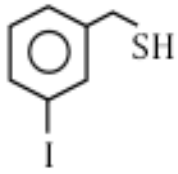
(3) $F^- > CH_3S^- > CH_3\text{–}COO^- > CH_3O^-$

(4) $CH_3O^- > F^- > CH_3\text{–}COO^- > CH_3S^-$

18) 

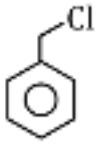
(1) 

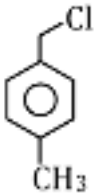
(2) 

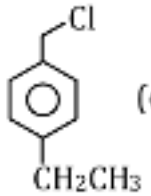
(3) 

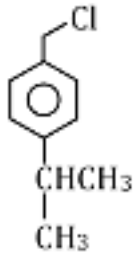
(4) None of these

19) Reactivity order for S_N1 reaction :-

(a) 

(b) 

(c) 

(d) 

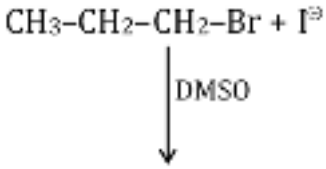
Watermarkly

(1) d > c > b > a

(2) a > b > c > d

(3) a > d > c > b

(4) b > c > d > a

20) 

This reaction would follow which of the following pathway predominantly ?

(1) S_N1

(2) S_N2

(3) E1

(4) E2

21) Which of following is not correctly match

(1) Column chromatography – Al_2O_3 is stationary phase

(2) Paper chromatography – Stationary phase is water

(3) Thin layer chromatography – Example of partition chromatography

(4) Chromatography can be use to check purity of compounds

22) Correct match :-

	Mixture		Purification technique
(A)	Chloroform & aniline	(P)	Distillation
(B)	Glycerol & Spentlye	(Q)	Chromatography
(C)	Aniline & Water	(R)	Steam distillation
(D)	Amino Acid	(S)	Distillation under reduced pressure

(1) A–P, B–Q, C–R, D–S

(2) A–P, B–S, C–R, D–Q

(3) A–P, B–R, C–S, D–Q

(4) A–R, B–S, C–P, D–Q

23) The purification method where solid substance change from vapour state without passing through the liquid state is called as

(1) Sublimation

(2) Crystallization

(3) Distillation

(4) Differential extraction

24) Chloroform (B.P. 334 K) and aniline (B.P. 457 K) are easily separated by the technique of

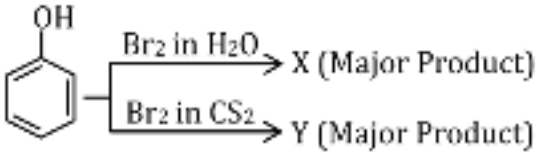
Watermarkly

(1) Crystallisation

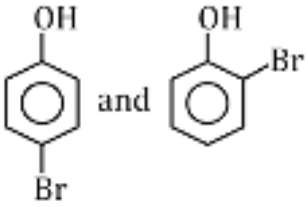
(2) Distillation

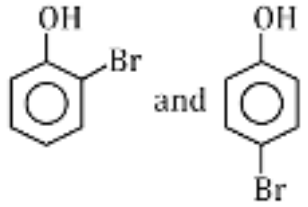
(3) Sublimation

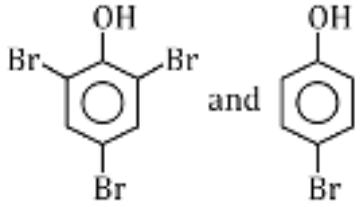
(4) Chromatography

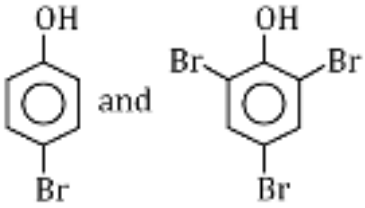
25) 

What is X and Y in above reaction respectively :-

(1) 

(2) 

(3) 

(4) 

26) The compound A, B and C in the reaction sequence

$CH_3—CH_2OH \xrightarrow{PBr_3} A \xrightarrow{Alc.\ KOH} B \xrightarrow{Br_2} C$

are given by the set :-

(1) C_2H_5Br, CH_3CH_2OH, CH_3CHBr_2

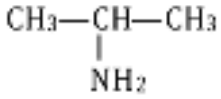
(2) C_2H_5Br, $CH{\equiv}CH$, $CH_2{=}CHBr$

(3) C_2H_5Br, $CH_2{=}CH_2$, $\overset{Br}{\underset{|}{CH_2}}-\overset{Br}{\underset{|}{CH_2}}$

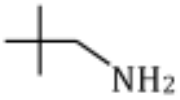
(4) C_2H_5Br, CH_3CH_2OH, $Br–CH_2–CH_2–Br$

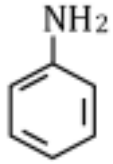
27) Which of the following will not give isocyanide test :-

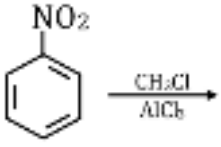
Watermarkly

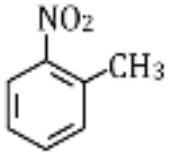
(1) 

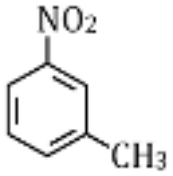
(2) CH_3–CH_2–NH–CH_3

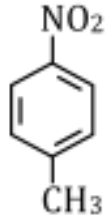
(3) 

(4) 

28)  A, A can be :-

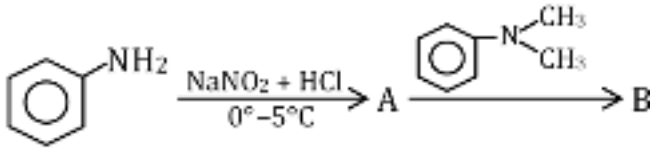
(1) 

(2) 

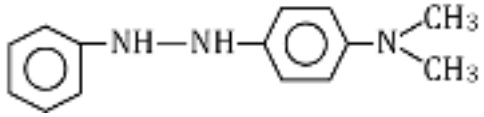
(3) 

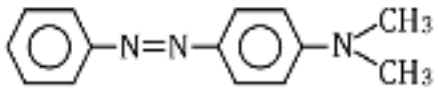
(4) No reaction

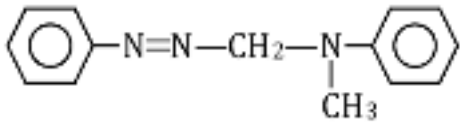
29) A coloured product is obtained in the following reaction of Aniline :-



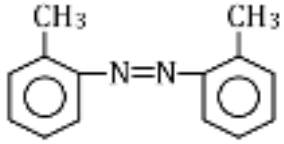
structure of 'B' is :-

(1) 

(2) 

(3) 

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(4) 

30) Lactose has :-

(1) α Glycosidic linkage between C_1 of galactose and C_4 of glucose

(2) β glycosidic linkage between C_1 of β D glucose and C_4 of β D galactose

(3) β Glycosidic linkage between C_1 of β D galactose and C_4 of β D glucose

(4) β glycosidic linkage between C_1 of galactose and C_2 of fructose

31) The purest form of vitamin 'C' is -

(1) Thiamine

(2) Riboflavin

(3) Ascorbic acid

(4) Pyridoxine

32) Match the column :

	Column-I(Vitamins)		**Column-II(Deficiency Disease)**
(a)	Vit A	(p)	Osteomalacia
(b)	Vit B12	(q)	Prenicious anaemia
(c)	Vit C	(r)	Xeropthalmia
(d)	Vit D	(s)	Scurvy

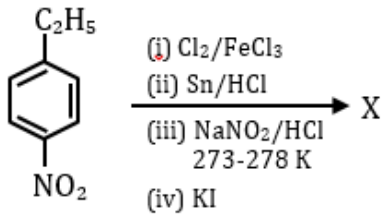
(1) a-p, b-q, c-r, d-s

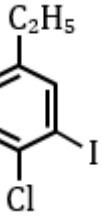
(2) a-r, b-p, c-s, d-q

(3) a-r, b-s, c-p, b-q

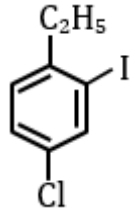
(4) a-r, b-q, c-s, d-p

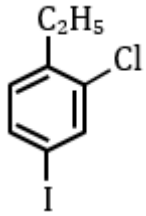
33) The major product X formed in the following reaction sequence is :

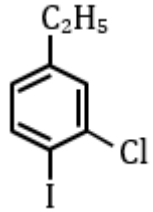


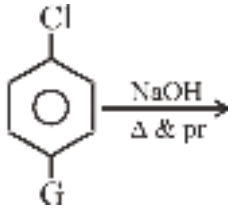
(1) 

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(2) 

(3) 

(4) 

34) 

:-

Rate of reaction is maximum if G is :-

(1) $-OCH_3$

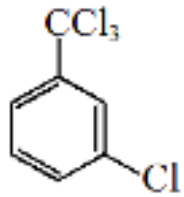
(2) $-CH_3$

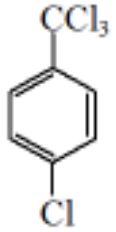
(3) $-NO_2$

(4) $-H$

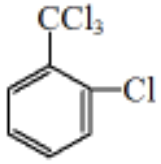
35) $C_6H_5CCl_3 \xrightarrow[Fe]{Cl_2} X$

In the above reaction X is

(1) 

(2) 

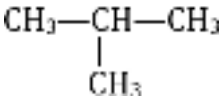
Watermarkly

(3) 

(4) All in equal amount

36) $CH_3—CH_2—CH_2—Cl + CH_3MgBr \rightarrow ?$

(1) $CH_3—CH_2—CH_2—CH_3$

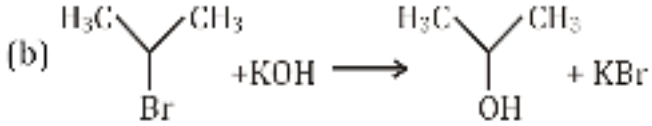
(2) 

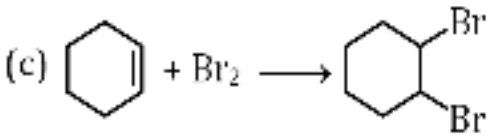
(3) $CH_3—CH_2—CH_2—CH_2—CH_2—CH_3$

(4) CH_4

37) For the following reactions :-

(a) $CH_3CH_2CH_2Br + KOH \rightarrow CH_3CH=CH_2 + KBr + H_2O$

(b) 

(c) 

Which of the following statement is **correct** ?

(1) (a) and (b) are elimination reaction and (c) is addition reaction

(2) (a) is elimination, (b) is substitution and (c) is addition reaction

(3) (a) is elimination, (b) and (c) are substitution reactions

(4) (a) is substitution, (b) and (c) are addition reaction

38) Elimination reaction of 2-Bromopentane to form pent-2-ene is:

(a) β-Elimination reaction

(b) Follow Zaitsev rule

(c) Dehydrohalogenation reaction

(d) Dehydration reaction

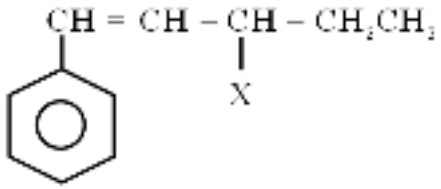
(1) (a), (b), (d)

(2) (a), (b), (c)

(3) (a), (c), (d)

(4) (b), (c), (d)

39) The given compound

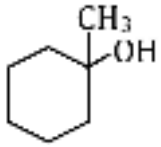


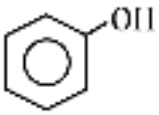
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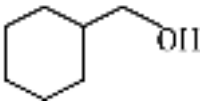
is an example of _____.

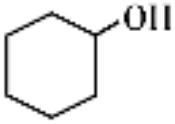
(1) aryl halide
(2) allylic halide
(3) vinylic halide
(4) benzylic halide

40) Which is most reactive with H-X ?

(1) 

(2) 

(3) 

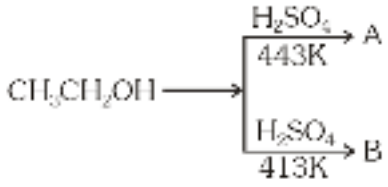
(4) 

41) The reaction



is called :-

(1) Schotten Bauman reaction
(2) Kolbe Schmidt reaction
(3) Reimer–Tiemann reaction
(4) Lederer–Manasse reaction

42)  A and B are (respectively)

(1) 

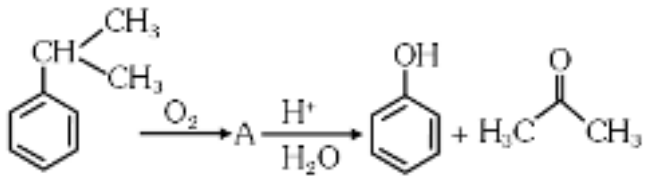
(2) 

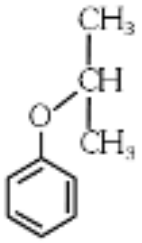
(3) $CH_2 = CH_2$, $CH_2 = CH_2$

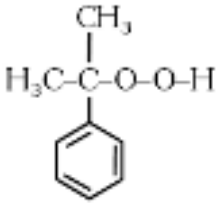
(4) 

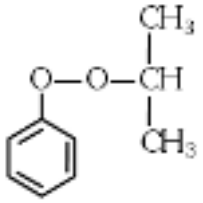
43) The structure of intermediate A in the following reaction is :-

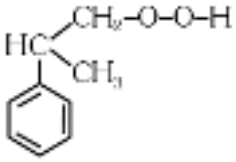
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(1) 

(2) 

(3) 

(4) 

44) The major products C and D formed in the following reactions respectively are :-

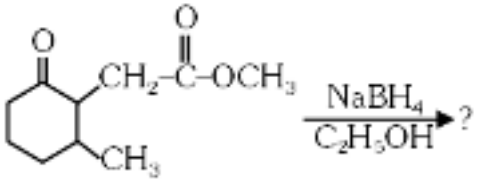
$H_3C–CH_2–CH_2–O–C(CH_3)_3 \xrightarrow[\Delta]{\text{excess HI}} C + D$

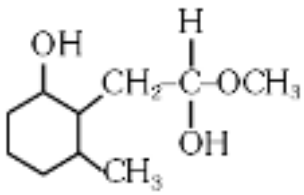
(1) $H_3C–CH_2–CH_2–I$ and $I–C(CH_3)_3$

(2) $H_3C–CH_2–CH_2–OH$ and $I–C(CH_3)_3$

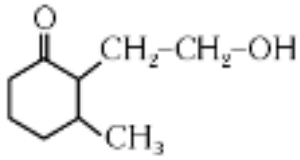
(3) $H_3C–CH_2–CH_2–I$ and $HO–C(CH_3)_3$

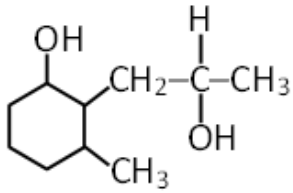
(4) $H_3C–CH_2–CH_2–OH$ and $HO–C(CH_3)_3$

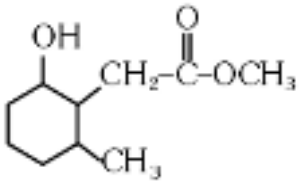
45) The product formed in the following chemical reaction is 

(1) 

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(2) 

(3) 

(4) 

BIOLOGY

1) Consider the following statements regarding DNA structure:
1. A nucleotide contains a nitrogenous base, pentose sugar and phosphate group.
2. Purines include adenine and guanine.
3. Uracil is present in DNA in place of thymine.
4. Two nucleotides are linked through a 3′–5′ phosphodiester linkage.
5. The backbone of a polynucleotide chain is formed by sugar and phosphates.
Which of the above statements are correct?

(1) 1, 2, 4 and 5 only

(2) 1, 2 and 3 only

(3) 2, 3, 4 and 5 only

(4) 1, 2, 3, 4 and 5

2) **Assertion (A):** DNA is chemically and structurally more stable than RNA.
Reason (R): DNA contains an additional –OH group at the 2′ position of ribose.

(1) Both A and R are true, and R is the correct explanation of A

(2) Both A and R are true, but R is not the correct explanation of A

(3) A is true, but R is false

(4) A is false, but R is true

3) Regarding eukaryotic transcription, which statements are correct?
1. RNA polymerase I transcribes 28S, 18S and 5.8S rRNAs.
2. RNA polymerase III transcribes tRNA, 5S rRNA and snRNAs.
3. RNA polymerase II transcribes hnRNA.
4. Primary transcripts contain only exons.
5. Tailing adds adenylate residues at the 3′ end.

(1) 1, 2 and 3 only

(2) 1, 2, 3 and 5 only

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(3) 2, 4 and 5 only

(4) All five

4) Match the following:

Column I	Column II
A. z gene	P. Permease
B. y gene	Q. Transacetylase
C. a gene	R. β-galactosidase
D. i gene	S. Repressor

(1) A–R, B–P, C–Q, D–S

(2) A–P, B–R, C–S, D–Q

(3) A–Q, B–P, C–R, D–S

(4) A–R, B–Q, C–P, D–S

5)

Match the following:

Column I	Column II
A. 5′ end of hnRNA	P. Adenylate residues
B. 3′ end of hnRNA	Q. Methyl guanosine triphosphate
C. Introns	R. Removed during splicing
D. Exons	S. Joined during splicing

(1) A–Q, B–P, C–R, D–S

(2) A–P, B–Q, C–S, D–R

(3) A–Q, B–P, C–S, D–R

(4) A–P, B–Q, C–R, D–S

6) Regarding DNA packaging, identify the correct statements:
1. DNA in the nucleoid of bacteria is organised in large loops held by proteins.
2. Histones are rich in lysine and arginine.
3. Histone octamer contains eight histone molecules.
4. A typical nucleosome contains about 200 bp of DNA helix.
5. Heterochromatin is transcriptionally active.

(1) 1, 2 and 3 only

(2) 1, 2, 3 and 4 only

(3) 2, 4 and 5 only

(4) All five

7) Arrange the following events in translation in their correct order:
A. Charged tRNA binds appropriate codon
B. Ribosome binds mRNA at AUG
C. Peptide bonds form

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D. Release factor binds stop codon
E. Complete polypeptide is released

(1) A → B → C → D → E
(2) B → A → C → D → E
(3) B → C → A → D → E
(4) B → A → D → C → E

8) Arrange the steps in HGP sequencing methodology:
A. DNA fragmentation
B. Cloning of fragments
C. Sequencing
D. Arrangement based on overlaps
E. Computer-assisted alignment
F. Annotation and chromosome assignment

(1) A → B → C → D → E → F
(2) B → A → C → D → F → E
(3) A → C → B → D → E → F
(4) A → B → D → C → E → F

9) Which sequence correctly represents the relationship between a eukaryotic gene and its mature RNA?

(1) Gene → only introns → mature RNA
(2) Gene → primary transcript containing exons + introns → splicing → mature RNA
(3) Gene → mature RNA without transcription
(4) Gene → only exons → splicing → introns

10) A DNA strand has the following sequence:
3′-ATGCATGC-5′
The complementary strand written in the **5′→3′ direction** is:

(1) 5′-TACGTACG-3′
(2) 5′-GCATGCAT-3′
(3) 5′-UACGUACG-3′
(4) 3′-TACGTACG-5′

11) Consider the following sequence of events:
I. Repressor binds operator
II. RNA polymerase is prevented from transcribing
III. Inducer interacts with repressor
IV. Repressor becomes inactive
V. RNA polymerase gains access to promoter
Which sequence represents switching **on** of lac operon after addition of lactose?

(1) I → II → III → IV → V
(2) III → IV → V

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(3) II → I → III → V → IV

(4) V → IV → III

12) Which of the following combinations contains only processes in which complementarity is explicitly important according to the chapter?

(1) Replication and transcription

(2) Only replication

(3) Only transcription

(4) Only translation

13) A mutation changes one base pair in the β-globin gene and changes glutamate to valine. This is an example of:

(1) Frameshift mutation

(2) Point mutation

(3) Chromosomal deletion

(4) Chromosomal rearrangement

14) Which of the following correctly distinguishes coding strand from template strand?

(1) Coding strand is 3′→5′ and template is 5′→3′

(2) Coding strand has the same sequence as RNA except thymine at place of uracil, whereas template strand is complementary to RNA

(3) Both strands act simultaneously as templates for a particular transcription unit

(4) Coding strand contains uracil instead of thymine

15) Which of the following combinations would produce a characteristic DNA fingerprint pattern?

(1) Identical repeat numbers of VNTR in all chromosomes

(2) Highly polymorphic VNTR regions with variable repeat copy numbers

(3) Only protein-coding regions

(4) Only mRNA sequences

16) Which statement best explains the term “nearly universal” for the genetic code?

(1) Every amino acid has exactly one codon

(2) The same codon generally specifies the same amino acid from bacteria to humans, although exceptions occur

(3) All organisms have exactly the same DNA sequence

(4) Stop codons are identical to amino-acid codons

17) A bacterial cell is synthesising a protein while the corresponding mRNA is still being transcribed. Which feature of bacteria permits this?

(1) Presence of introns

(2) Extensive RNA processing

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(3) Absence of separation between transcription and translation compartments

(4) Presence of three nuclear RNA polymerases

18) A scientist wants to determine DNA-level variation between individuals without sequencing the entire genome. Which sequence of reasoning is most appropriate?

(1) Repetitive DNA → polymorphism → VNTR → DNA fingerprinting

(2) Protein → tRNA → ribosome → DNA fingerprinting

(3) Exon → intron → splicing → fingerprinting

(4) mRNA → codon → anticodon → fingerprinting

19) A double-stranded DNA molecule contains 20% cytosine. What percentage of adenine will it contain?

(1) 20%

(2) 30%

(3) 40%

(4) 60%

20) The DNA of a mammalian cell is approximately **6.6×10^9 bp** and the distance between consecutive base pairs is **0.34 nm**. Its approximate total DNA length is:

(1) 0.22 m

(2) 2.2 m

(3) 22 m

(4) 220 m

21) An mRNA contains the sequence:
5′-AUG UUU UUC UAG-3′
How many amino acids will be incorporated before termination?

(1) 1

(2) 2

(3) 3

(4) 4

22) **Assertion (A):** Euchromatin is generally transcriptionally active.
Reason (R): Euchromatin is tightly packed and stains lightly.

(1) Both the Assertion (A) and Reason (R) are true, and the Reason is the correct explanation of the Assertion.

(2) Both the Assertion (A) and Reason (R) are true, but the Reason is not the correct explanation of the Assertion.

(3) The Assertion (A) is true, but the Reason (R) is false.

(4) The Assertion (A) is false, but the Reason (R) is true.

23) **Assertion (A):** In bacteria, transcription and translation can be coupled.

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Reason (R): Bacterial mRNA does not require processing to become active and transcription and translation occur in the same compartment.

(1) Both the Assertion (A) and Reason (R) are true, and the Reason is the correct explanation of the Assertion.

(2) Both the Assertion (A) and Reason (R) are true, but the Reason is not the correct explanation of the Assertion.

(3) The Assertion (A) is true, but the Reason (R) is false.

(4) The Assertion (A) is false, but the Reason (R) is true.

24) **Assertion (A):** The coding strand is used as the template for RNA synthesis.
Reason (R): The coding strand has 5′→3′ polarity and has the same sequence as RNA except for T/U difference.

(1) Both the Assertion (A) and Reason (R) are true, and the Reason is the correct explanation of the Assertion.

(2) Both the Assertion (A) and Reason (R) are true, but the Reason is not the correct explanation of the Assertion.

(3) The Assertion (A) is true, but the Reason (R) is false.

(4) The Assertion (A) is false, but the Reason (R) is true.

25) In the Avery-MacLeod-McCarty experiment, transformation was inhibited by:

(1) Protease

(2) RNase

(3) DNase

(4) Lipase

26) Hershey and Chase used radioactive phosphorus because:

(1) DNA contains phosphorus but protein does not

(2) Protein contains phosphorus but DNA does not

(3) Both DNA and protein contain phosphorus

(4) Phosphorus is present only in RNA

27) In prokaryotes such as E. coli, DNA is organised in a region called:

(1) Nucleus

(2) Nucleoid

(3) Nucleosome

(4) Chromatin

28) Histones are rich in which amino acid residues?

(1) Glycine and alanine

(2) Lysine and arginine

(3) Valine and leucine

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(4) Aspartate and glutamate

29) Which of the following statements about the DNA double helix is correct?

(1) Both strands have the same polarity
(2) The two strands are antiparallel
(3) Bases are present outside the helix
(4) Sugar-phosphate groups project inside

30) The S strain of *Streptococcus pneumoniae* differs from the R strain because S strain:

(1) Has no genetic material
(2) Has a mucous polysaccharide coat
(3) Lacks a cell wall
(4) Is non-virulent

31) A nucleosome consists of DNA wrapped around:

(1) A histone monomer
(2) A histone octamer
(3) Two histone molecules
(4) RNA molecules

32) A typical nucleosome contains approximately:

(1) 20 bp DNA
(2) 50 bp DNA
(3) 100 bp DNA
(4) 200 bp DNA

33)

Match the following HGP observations:

Column I	**Column II**
A. Chromosome 1	P. Fewest genes
B. Y chromosome	Q. Most genes
C. SNPs	R. Single-base differences
D. Repetitive DNA	S. Repeated many times

(1) A–Q, B–P, C–R, D–S
(2) A–P, B–Q, C–R, D–S
(3) A–Q, B–R, C–P, D–S
(4) A–S, B–P, C–R, D–Q

34) Which one of the following is a purine?

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(1) Cytosine

(2) Thymine

(3) Adenine

(4) Uracil

35) Which of the following statement is/are incorrect ?
A. The presence of introns is reminiscent of antiquity and the process or splicing represents the dominance of RNA world.
B. Marshall Nirenberg's cell-free system of protein synthesis finally helped the code to be deciphered.
C. There are at least three type of RNA polymerase enzymes in prokaryotes for transcription.

(1) Only B

(2) Only C

(3) Only A and B

(4) Only A

36)

Match the following biotechnology research areas with their correct functions.

Column I	**Column II**
A. Best catalyst	P. Purification of desired product
B. Optimal conditions	Q. Improved organism or pure enzyme
C. Downstream processing	R. Engineering conditions for catalyst action

Choose the correct option:

(1) A–Q, B–R, C–P

(2) A–P, B–Q, C–R

(3) A–R, B–P, C–Q

(4) A–Q, B–P, C–R

37) A tissue culture medium contains sucrose, inorganic salts, vitamins, amino acids, auxins and cytokinins. If sucrose is removed, which essential role is directly affected?

(1) DNA replication

(2) Carbon source

(3) Chromosome doubling

(4) Cell wall digestion

38) **Assertion (A):** Plants produced through micropropagation are generally genetically identical to the original plant.
Reason (R): Micropropagation involves regeneration of plants through tissue culture from an explant.

(1) Both A and R are true, and R correctly explains A

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(2) Both A and R are true, but R does not correctly explain A

(3) A is true, R is false

(4) A is false, R is true

39) The famous "pomato" mentioned in NCERT is produced by fusion of:

(1) Tomato ovum and potato sperm

(2) Tomato protoplast and potato protoplast

(3) Tomato pollen and potato ovule

(4) Tomato seed and potato seed

40) Arrange the following events in the correct sequence after a susceptible insect consumes Bt toxin:
A. Activation of toxin
B. Solubilisation of crystal
C. Binding to midgut epithelial cells
D. Pore formation
E. Cell swelling and lysis

(1) B → A → C → D → E

(2) A → B → C → E → D

(3) C → B → A → D → E

(4) B → C → A → D → E

41) Which pair of *Bt* genes is associated with control of cotton bollworms according to NCERT?

(1) *cryIAb* and *cryIIAc*

(2) *cryIAc* and *cryIIAb*

(3) *cryIAb* and *cryIAc*

(4) *cryIIAb* and *cryIAb* only

42) A transgenic tobacco plant is protected against *Meloidogyne incognita*. The underlying mechanism involves:

(1) DNA methylation only

(2) RNA interference

(3) Somatic hybridisation

(4) Micropropagation

43) **Assertion (A):** Meristem culture can be used to obtain virus-free plants.
Reason (R): The meristem is free of virus even when the plant is infected

(1) Both A and R are true, and R is the correct explanation of A.

(2) Both A and R are true, but R is not the correct explanation of A.

(3) A is true, R is false.

(4) A is false, R is true.

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44) A student claims that the C-peptide is a component of mature insulin. Which statement correctly evaluates this claim?

(1) Correct, because C-peptide connects A and B chains in mature insulin
(2) Incorrect, because C-peptide is removed during maturation
(3) Correct, because C-peptide is responsible for insulin receptor binding
(4) Incorrect, because insulin has only one polypeptide chain

45) Gene therapy is best defined as:

(1) Treatment of every disease using antibiotics
(2) Correction of a genetic defect by introducing a normal functional gene
(3) Production of insulin from pancreas
(4) Destruction of all abnormal cells using antibodies

46) The first clinical gene therapy was given in:

(1) 1975
(2) 1983
(3) 1990
(4) 2000

47) PCR can detect very low concentrations of a pathogen because it:

(1) Amplifies the pathogen's nucleic acid
(2) Produces antibodies directly
(3) Destroys the pathogen's cell wall
(4) Converts proteins into DNA

48) Which of the following sequences represents the correct principle of ELISA?

(1) Antigen–antibody interaction → detection of infection
(2) DNA amplification → detection of infection
(3) Protoplast fusion → detection of infection
(4) Disulphide bridge formation → detection of infection

49) Which pair is **incorrectly** matched?

(1) ADA deficiency — Gene therapy
(2) Early diagnosis — PCR
(3) Pathogen antigen detection — ELISA
(4) Insulin production — RNA interference

50) A researcher introduces a foreign gene into a mouse. The gene modifies the formation of insulin-like growth factor. The researcher then studies changes in growth and development.
Which inference is most appropriate?

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(1) The mouse is being used as a model for vaccine safety.

(2) The mouse is being used to study the biological role of a growth-related factor.

(3) The mouse is being used to produce recombinant insulin.

(4) The mouse is being used to study RNA interference.

51) Match the Following

Column I	Column II
A. Rosie	P. First transgenic cow
B. Alpha-1-antitrypsin	Q. Treatment of emphysema
C. Alpha-lactalbumin	R. Human protein-enriched milk
D. Insulin-like growth factor	S. Study of normal growth

(1) A-P, B-Q, C-R, D-S

(2) A-Q, B-R, C-S, D-P

(3) A-R, B-P, C-Q, D-S

(4) A-P, B-R, C-S, D-Q

52) A. Rosie produced human alpha-lactalbumin-enriched milk.
B. Transgenic mice are used for polio vaccine safety testing.
C. Transgenic animals can serve as models for cancer.
D. GEAC is involved in the approval of all conventional breeding programmes.
Which statements are correct?

(1) A and B only

(2) A, B and C only

(3) B, C and D only

(4) A, B, C and D

53) Which of the following represents a correct sequence of biotechnology applications?

(1) Foreign gene introduction → Transgenic animal → Biological product production

(2) Foreign gene introduction → Somaclone formation → PCR

(3) Foreign gene introduction → Protoplast fusion → ELISA

(4) Foreign gene introduction → Meristem culture → Vaccine testing

54) Consider the following statements about biopiracy:
A. It may involve multinational companies.
B. It can involve traditional knowledge.
C. It always requires proper authorisation.
D. It may involve exploitation without compensatory payment.
Choose the correct option:

(1) A and B only

(2) A, B and D only

(3) B and C only

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(4) A, B, C and D

55) Consider the following statements:
A. Genetic modification may have unpredictable effects when modified organisms are introduced into ecosystems.
B. GEAC is associated with decisions regarding the validity of GM research and safety of introducing GM organisms for public services.
C. Ethical issues are limited only to the safety of human beings.
D. Patenting biological resources and traditional knowledge can create ethical concerns.
Choose the correct option:

(1) A and B only
(2) A, B and D only
(3) B, C and D only
(4) A, B, C and D

56)

I. The bacterial cells must be made ___**A**___ to take up DNA.
II. The cell is treated with specific concentration of a divalent cation such as ___**B**___ to increase the efficiency with which DNA enters.
III. In ___**C**___ method recombinant DNA is directly injected into the nucleus of an animal cell.

The most appropriate option regarding A, B and C is :

	A	**B**	**C**
(1)	Competent	Calcium	Gene gun method
(2)	Transformation	Sodium	Microinjection method
(3)	Competent	Calcium	Microinjection method
(4)	Transformation	Sodium	Gene gun method

(1) 1
(2) 2
(3) 3
(4) 4

57) Which of the following statement is incorrect with respect to plasmids ?

(1) Stanley Cohen and Herbert Boyer in 1972 constructed the first recombinant DNA using a native plasmid of *E. Coli.*
(2) Plasmid is an autonomously replicating circular extra-chromosomal DNA.
(3) Plasmid DNA act as vectors to transfer the piece of DNA attached to it.

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(4) The linking of antibiotic resistance gene with the plasmid vector become possible with help of DNA ligase.

58) Arrange the following steps in the correct sequence involved in the construction of a recombinant DNA molecule, as demonstrated by Cohen and Boyer.
1. Insertion of the antibiotic resistance gene into the plasmid.
2. Isolation of the plasmid from *Salmonella typhimurium*.
3. Cutting the plasmid and the antibiotic resistance gene using restriction enzymes.
4. Introduction of recombinant plasmid into a host bacterium.
5. Selection of transformed cells using antibiotics.

(1) 2 → 3 → 1 → 4 → 5

(2) 1 → 2 → 3 → 4 → 5

(3) 2 → 1 → 3 → 4 → 5

(4) 3 → 2 → 1 → 5 → 4

59)

Match the items in Column A with the correct descriptions in Column B and choose the correct option.

Column A	**Column B**
A. Restriction endonuclease	1. Joins DNA fragments
B. DNA ligase	2. Adds methyl groups to DNA
C. Methylase	3. Cuts DNA at specific recognition sequences
D. Plasmid	4. Circular DNA vector used in genetic engineering
E. Recognition sequence	5. Specific DNA sequence recognized by restriction enzymes

(1) A-3, B-1, C-2, D-4, E-5

(2) A-2, B-3, C-1, D-5, E-4

(3) A-1, B-2, C-3, D-5, E-4

(4) A-4, B-5, C-2, D-3, E-1

60) A foreign DNA is inserted into the BamHI site of the tetracycline resistance gene in pBR322. The bacteria are then grown on ampicillin-containing medium. Which of the following observations is most likely?

(1) Only non-recombinant cells will grow

(2) Only recombinant cells will grow

(3) Both recombinant and non-recombinant cells will grow

(4) No cells will grow due to insertional inactivation

61) Consider the following statements regarding PCR:
Statement I: DNA polymerase, which remain active during the high temperature induced denaturation of double stranded DNA.
Statement II: The extension step in PCR occurs at around 55°C, which is the optimal temperature for DNA polymerase activity.

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Which of the following is correct?

(1) Both statements I and II are correct
(2) Statement I is correct, but Statement II is incorrect
(3) Statement I is incorrect, but Statement II is correct
(4) Both statements I and II are incorrect

62) Consider the following statements related to large-scale recombinant protein production:
Statement I: Bioreactors are designed to provide optimal growth conditions for microbial or cell cultures, which leads to efficient production of desired proteins.
Statement II: Small-scale cultures in test tubes are equally efficient as bioreactors in yielding large quantities of recombinant proteins.
Which of the following is correct?

(1) Both statements I and II are correct
(2) Statement I is correct, but Statement II is incorrect
(3) Statement I is incorrect, but Statement II is correct
(4) Both statements I and II are incorrect

63) **Statement I:** Downstream processing involves clinical trials and product formulation.
Statement II: Clinical trials are optional and only used in agricultural biotechnology.

(1) Both statements are true, and Statement II explains Statement I
(2) Both statements are true, but Statement II does not explain Statement I
(3) Statement I is true, but Statement II is false
(4) Both statements are false

64) Why is Taq polymerase used in PCR instead of human DNA polymerase?

(1) It synthesizes longer DNA fragments
(2) It is more accurate
(3) It is stable at high temperatures
(4) It has proofreading ability

65) A recombinant plasmid containing a gene of interest was inserted within the lacZ gene. Which of the following is a correct implication of this?

(1) The host bacteria will degrade the recombinant plasmid
(2) The recombinant bacteria will appear blue in the presence of X-gal
(3) The lacZ gene is disrupted, so β-galactosidase will not be produced
(4) All transformed cells will fluoresce under UV light

66) In gel electrophoresis the DNA fragments separate according to the size (smaller the fragment size, the faster it moves) this is called :-

(1) Sieving effect
(2) Movement effect

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(3) Spooling

(4) Density effect

67) Why is it preferable for a cloning vector to have a single unique recognition site for a restriction enzyme?

(1) It prevents the vector from replicating in the host

(2) Multiple restriction sites can cause the vector to be cut in to multiple fragments

(3) It increases the antibiotic resistance

(4) It enhances the activity of DNA polymerase

68) During recombinant DNA preparation, purified DNA and vector DNA are both digested with the same restriction enzyme before ligation. Which of the following statements about this process is correct?

(1) DNA moves towards the cathode in agarose gel because it is negatively charged

(2) Restriction enzymes cut DNA at random sequences, creating sticky ends.

(3) Using the same restriction enzyme on vector and insert generates compatible sticky ends for ligation.

(4) DNA ligase cuts the DNA at restriction sites to allow insertion of the gene of interest.

69) For transformation, microparticles coated with DNA to be bombarded with gene gun are made up of :

(1) Platinum or Zinc

(2) Gold or Tungsten

(3) Silicon or Platinum

(4) Silver or Platinum

70) The Ti plasmid of *Agrobacterium tumefaciens* is used as a vector in plant genetic engineering because:

(1) It kills plant cells

(2) It integrates its T-DNA into the plant genome

(3) It improves chlorophyll synthesis

(4) It enhances flower development

71) **Assertion (A):** According to the European Federation of Biotechnology (EFB), biotechnology is the integration of natural science and organisms, cells, parts there of and molecular analogues for products and services.
Reason (R): Genetic engineering is the technique to alter the chemistry of genetic material.

(1) Both A and R are true, and R is the correct explanation of A.

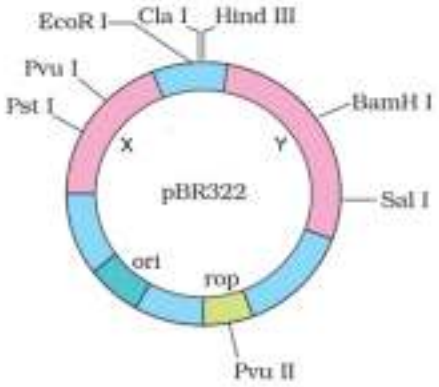
(2) Both A and R are true, but R is not the correct explanation of A.

(3) A is true, but R is false.

(4) A is false, but R is true.

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72) What is the significance of X and Y in pBR322?



(1) They allow the vector to replicate independently in the host

(2) They enable the selection of recombinants

(3) They are used to degrade the plasmid DNA

(4) They help in the expression of foreign proteins

73) What does the white thread-like structure formed upon adding ethanol, shown in the diagram indicate?



(1) Protein precipitation

(2) RNA formation

(3) DNA precipitation

(4) Lipid separation

74) Lactic acid bacteria convert milk into curd and improves its nutritional quality by enhancing

(1) Vitamin A

(2) Vitamin B

(3) Vitamin C

(4) Vitamin D

75) Streptokinase, used as a 'clot buster' is obtained from

(1) *Streptococcus*

(2) *Staphylococcus*

(3) *Lactobacillus*

(4) *Saccharomyces*

76) Which bacterium helps in the production of 'Swiss cheese'?

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(1) *Propionibacterium sharmanii*

(2) *Trichoderma polysporum*

(3) *Saccharomyces cerevisiae*

(4) *Aspergillus niger*

77) *Monascus purpureus* is a yeast used commercially in the production of

(1) Ethanol

(2) Streptokinase for removing clots from the blood vessels

(3) Citric acid

(4) Blood cholesterol lowering statins

78) Microbe used for biocontrol of pest butterfly caterpillars is :

(1) *Trichoderma sp.*

(2) *Saccharomyces cerevisiae*

(3) *Bacillus thuringiensis*

(4) *Streptococcus sp.*

79) **Assertion :** Biogas has been used as a fuel for the purposes of lighting and cooking.
Reason : This technology was developed in India due to efforts of IARI and KVDC.

(1) Both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

80)

The given table contains type of microbe (Column I), Scientific name (Column II) and commercial product (Column III). Some names are replaced by A, B, C and D.
Identify the correct names.

Type of Microbe	**Scientific Name**	**Commercial Product**
Bacterium	**A**	Lactic acid
Fungus	**B**	Cyclosporin A
C	*Monascus purpureus*	Statins
Fungus	*Penicillium notatum*	**D**

(1) A – *Lactobacillus*, B – *Trichoderma polysporum*, C – Yeast (fungus), D – Penicillin

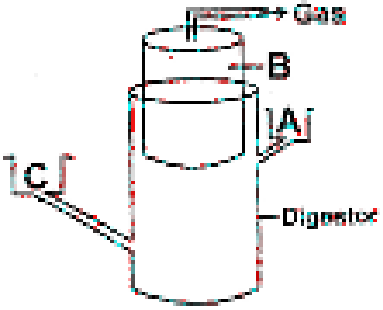
(2) A – *Lactobacillus*, B – *Trichoderma polysporum*, C – Yeast (algae), D – Penicillin

(3) A – *Lactobacillus*, B – *Trichoderma polysporum*, C – Yeast (prokaryote), D – Penicillin

(4) A – *Lactobacillus*, B – *Trichoderma polysporum*, C – *Agaricus* (fungus), D – Penicillin

81) The diagram below shows a typical biogas plant. With few structure labelled as A, B and C.
Identify A, B and C.

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(1) A – Sludge, B – Methane, Oxygen, C – Dung, water

(2) A – Sludge, B – Methane, Carbon dioxide, C– Dung, water

(3) A – Sludge, B – Ethylin, Carbon dioxide, C – Dung, water

(4) A – Sludge, B – Methane, Carbon dioxide, C – Sewage

82) **Assertion :** Biopesticides help in overcoming the disadvantages of chemical pesticides.
Reason : Microorganisms used as biopesticide include viruses, bacteria, and protozoa.

(1) Both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

83) Symbiotic association is exhibited by
I. Mycorrhiza
II. *Rhizobium*
III. Heterocyst
IV. Yeast

(1) I, III, IV

(2) I, II, III, IV

(3) II, III, IV

(4) I, II

84) During sewage treatment, biogases produced include which of the following gases ?

(1) Methane, oxygen, hydrogen sulphide

(2) Hydrogen sulphide, methane, sulphur dioxide

(3) Hydrogen sulphide, nitrogen, methane

(4) Methane, hydrogen sulphide, carbon dioxide

85) Primary sludge is used for the

(1) Preparation of compost

(2) Preparation of manure

(3) Biogas production

(4) All of these

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86) The chemical substances produced by some microbes which can kill or retard the growth of other microbes are called _____

(1) Toddy
(2) Lactic acid
(3) Antibiotics
(4) Ethanol

87) Which one of the following is used in the manufacture of alcohol ?

(1) Bacteria
(2) Bread molds
(3) Yeasts
(4) Slime molds

88) Restriction enzyme EcoR I cuts the DNA between bases G and A only when the sequence in DNA is :

(1) GATATC
(2) GAATTC
(3) GATTCC
(4) GAACTT

89) Upon exposure to UV radiation, DNA stained with ethidium bromide will show

(1) Bright blue colour
(2) Bright yellow colour
(3) Bright orange colour
(4) Bright red colour

90)

Match the following organisms with the products they produce :-

(a)	*Lactobacillus*	(i)	Cheese
(b)	*Saccharomyces cerevisiae*	(ii)	Curd
(c)	*Aspergillus niger*	(iii)	Citric Acid
(d)	*Acetobacter aceti*	(iv)	Bread
		(v)	Acetic Acid

Select the **correct** option.

	(a)	**(b)**	**(c)**	**(d)**
(1)	(ii)	(iv)	(v)	(iii)
(2)	(ii)	(iv)	(iii)	(v)
(3)	(iii)	(iv)	(v)	(i)

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(4)	(ii)	(i)	(iii)	(v)

(1) 1

(2) 2

(3) 3

(4) 4

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ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	3	1	2	2	1	3	3	3	3	3	1	4	4	3	3	3	3	2	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	4	4	2	1	3	2	2	2	2	1	3	1	1	4	4	4	4	2	3	1
Q.	41	42	43	44	45															
A.	3	4	3	1	4															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	2	2	4	4	3	3	3	4	3	3	2	2	4	2	1	1	3	4	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	2	1	2	3	3	2	4	2	3	3	4	3	3	1	1	2	2	2	1
Q.	86	87	88	89	90															
A.	2	4	2	1	4															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	1	3	2	1	1	2	2	1	2	1	2	1	2	2	2	2	3	1	2	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	2	1	4	3	1	2	2	2	2	2	4	1	3	2	1	2	2	2	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	2	1	2	2	3	1	1	4	2	1	2	1	2	2	3	1	1	1	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	2	2	3	3	3	1	2	3	2	2	2	2	3	2	1	1	4	3	2	1
Q.	171	172	173	174	175	176	177	178	179	180										
A.	2	2	4	4	4	3	3	2	3	2										

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SOLUTIONS

PHYSICS

1) **Explaining :** Find no. of collisions between the balls.
Concept : This question is based on elastic collision.
Solution :
After first collision ball 1 come to rest and ball 2 moves forward and as M < m after second collision the velocity of second ball will remain in same direction so no more collision will occur.
Final Answer : (2)

2) **Explaining :** Find initial velocity of shell.
Concept : This question is based on conservation of linear momentum (Gun-bullet system).



Solution :
COLM : $0 = -4v + 0.2u \Rightarrow v = \frac{0.2u}{4}$
Total K.E : $\frac{1}{2} \times 4v^2 + \frac{1}{2}(0.2)\,u^2 = 1.05 \times 1000$
$2\left(\frac{0.2u}{4}\right)^2 + 0.1u^2 = 0.105u^2 = 1.05 \times 1000$
u = 100 m/s
Final Answer : (3)

3) **Explaining :** Find value of x
Concept : This question is based on collision
Solution :
There is no external force on system (ball A and ball B). The balls moves due to mutual attraction that is internal force to the system. The centre of mass of the system will not move. They will collide at the position of centre of mass of ball A and B.
$m_A = 2P \times V \quad m_B = PV$
$\therefore m_A = 2m_B$
$x_{cm} = \frac{m_A \times A + m_B \times B}{m_A + m_B} = \frac{m_A.0 + m_B.10R}{(m_A + m_B)} = \frac{m_B \times 10R}{(2m_B + m_B)} = \frac{10}{3}R$
Final Answer : (1)

4) **Explaining :** Find the position of centre of mass at t = 2s
Concept : This question is based on centre of mass of discrete particles system.
Solution :
As no external force are acting on the system. The velocity of centre of mass will be constant.
$v_{cm} = \frac{1 \times 5 - 1 \times 3}{1 + 1} = 1$ m/s
Displacement of centre of mass $\Delta x_{cm} = 1 \times 2 = 2$ m.

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Hence position of centre of mass $\Delta x_{cm} = 5 + 2 = 7$ m

Final Answer : (2)

5) **Explaining :** find mass of second body

Concept : This question is based on elastic collision

Solution :



$$\vec{v}_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)\vec{u}_1 + \frac{2\vec{u}_2 m_2}{m_1 + m_2}$$

$$\frac{-u}{3} = \left(\frac{2-m}{2+m}\right)u$$

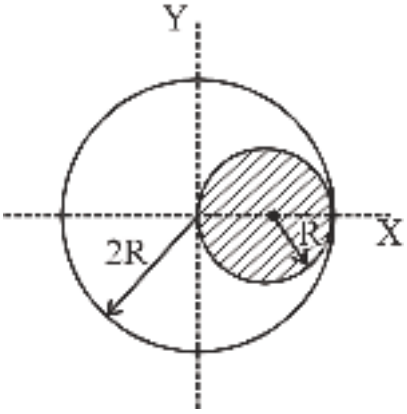
$$\frac{-1}{3} = \left(\frac{2-m}{2+m}\right) \Rightarrow m = 4kg$$

Final Answer : (2)

6) **Explaining :** Find the value of α.

Concept : This question is based on centre of mass of truncated body.

Solution :



$$x_{CM} = \frac{Mx - mx_1}{M - m}$$

$x = 0$, $x_1 = R$

$$m = \frac{M}{4\pi R^2} \times \pi R^2 = \frac{M}{4}$$

$$x_{CM} = \frac{M \times 0 - \frac{M}{4} \times R}{M - \frac{M}{4}} = -\frac{R}{3}$$

$$\alpha R = \frac{R}{3} \Rightarrow \alpha = \frac{1}{3}$$

Final Answer : (1)

7) **Explaining :** Find loss of kinetic energy of system in collision.

Concept : This question is based on perfectly inelastic collision.

Solution :

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loss in K.E. $= \frac{m_1 m_2}{2(m_1+m_2)}\left(1-e^2\right)(u_1-u_2)^2$

$= \frac{4\times 6}{2(4+6)}(1-0)(12-0)^2 = 172.8J$

Final Answer : (3)

8)

Question Explanation : A head-on elastic collision takes place between two identical balls in one dimension, find velocities after collision.
Concept : Head-on elastic collision in 1-D.
Solution :
In the case of a head-on elastic collision in 1-D, if the colliding bodies are identical, then their velocities get exchanged.
So,
$v_A = u_B = -0.3$ m/s
$v_B = u_A = 0.5$ m/s
Final Answer : Option (3)

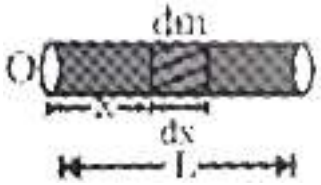
9) **Question Explanation :**
A 50 kg man in gravity-free space at 10 m height throws a 0.5 kg stone downward with 2 m/s speed. We need to find the man's final height when the stone reaches the floor.

Concept :
Conservation of Linear Momentum

Solution :

By conservation of linear momentum $m_{man}\, v_{man} = m_{stone}\, v_{stone}$ $50v_{man} = 0.5 \times 2$ $v_{man} = \frac{1}{50} = 0.02$ m/s (upward) Time taken for stone to reach floor: $t = \frac{10}{2} = 5\,s$ Man's displacement $s = v_{man} \times t = 0.02 \times 5 = 0.1$ m Final height of the man: 10 + 0.1 = 10.1 m **Final Answer :**
The man will be **10.1 m** above the floor **(Option-3).**

10) **Explaining :** Find the distance of the COM of rod from given point.
Concept : This question is based on COM of continous system.
Solution :



$dm = \mu dx = \lambda x^2 dx$

$x_{CM} = \frac{\int x\,dm}{\int dm} = \frac{\int_0^L x\lambda x^2 dx}{\int_0^L \lambda x^2 dx}$

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$$= \frac{\left|\frac{x^4}{4}\right|_0^L}{\left|\frac{x^3}{3}\right|_0^L} = \frac{\frac{L^4}{4}}{\frac{L^3}{3}} = \frac{3L}{4}$$

Final Answer : (3)

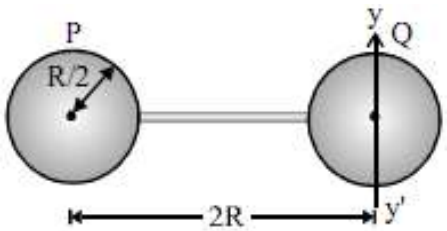
11) **Explaining :** Find coordinate of COM of system.
Concept : This question is based on centre of mass of composite bodies.
Solution : Coordinate COM of semicircular ring is $\left(0, -\frac{2R}{\pi}\right)$ = (0, –18)
& coordinate COM of semicircular disc is $\left(0, \frac{4R}{3\pi}\right)$ = (0, 12)
Coordinate of COM of system

$$Y_{COM} = \frac{M_1y_1 + M_2y_2}{M_1 + M_2}$$

$$= \frac{M(12) + M(-18)}{M + M}$$

Y_{COM} = –3 & X_{COM} = 0
Final Answer : (3)

12) **Explaining :** Find the MOI of system.
Concept : This question is based on MOI of composite system.
Solution :



Moment of inertia of the system about yy' ,
$I_{yy'}$ = Moment of inertia of sphere P about yy' + Moment of inertia of sphere Q about yy'
⇒ Moment of inertia of sphere P about yy' $= \frac{2}{5}M\left(\frac{R}{2}\right)^2 + M(x)^2$ [Parallel axis theorem]

$$= \frac{2}{5}M\left(\frac{R}{2}\right)^2 + M(2R)^2 = \frac{MR^2}{10} + 4MR^2$$

Moment of inertia of sphere Q about yy' is $= \frac{2}{5}M\left(\frac{R}{2}\right)^2$

Now $I_{yy'} = \frac{MR^2}{10} + 4MR^2 + \frac{2}{5}\left(\frac{R}{2}\right)^2$

$$= \frac{21}{5}MR^2$$

Final Answer : (1)

13) **Explaining :** We need to find radius of gyration.
Concept : This question is based on $K = \sqrt{\frac{I}{m}}$ (radius of gyration).
Solution : $160 = 10k^2 \Rightarrow k = 4$ m

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Final Answer : (4)

14) **Explaining :** Find the value of force.
Concept : This question is based on torque.
Solution : $\tau = 1.6 \times 1 = 1.6$ N-m, $F = \frac{\tau}{d} = \frac{1.6}{0.4} = 4N$
Final Answer : (4)

15) **Explaining :** Find velocity of ball.
Concept : This question is based on conservation of angular momentum.
Solution : Accoding to angular momentum conservation.
$mv_0L = I_{sys}\,\omega$

$$mv^{0}L = \left(mL^2 + \frac{mL^2}{3}\right)\omega$$

$$\omega = \frac{3v_0}{4L}$$

From energy conservation after collision

$$\frac{1}{2}I_{sys}\omega^2 = (mgL + mgL/2)$$

$$\frac{1}{2}\left(\frac{4}{3}mL^2\right)\left(\frac{3}{4}\frac{v_0}{L}\right)^2 = \frac{3}{2}mgL$$

$$\frac{mv_0^2}{4} = mgL \Rightarrow v_0 = 2\sqrt{gL}$$

Final Answer : (3)

16) **Explaining :** Find ratio of rotational KE to the total kinetic energy.
Concept : This question is based on energy in rolling motion.
Solution : $\frac{KE_{rotational}}{KE_{total}} = \frac{\frac{K^2}{R^2}}{1+\frac{K^2}{R^2}} = \frac{\frac{2}{5}}{1+\frac{2}{5}} = \frac{\frac{2}{5}}{\frac{7}{5}} = \frac{2}{7}$
Final Answer : (3)

17) **Explaining :** Find acceleration of free end of the spring.
Concept : This question is based on rotating pulley.
Solution : $FR = \tau = I\alpha$

$$FR = \frac{1}{2}mR^2\left(\frac{a}{R}\right) \text{ or } a = \frac{2F}{m}$$

Final Answer : (3)

18) **Explaining :** Find angular momentum about the axis of rotation.
Concept : This question is based on conservation of angular momentum.
Solution : As $\vec{\tau}_{ext} = 0$, so angular momentum will be conserved.
L = constant
Final Answer : (3)

19) **Explaining :** Find angular momentum.

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Concept : This question is based on angular momentum.
Solution : $I = mr^2 = 10 \times (0.2)^2 = 0.4$ kg-m^2

$\omega = 2\pi n = 2\pi \times \frac{2100}{60}$ rad/s

$\therefore L = I\omega = \frac{0.4 \times 2\pi \times 210}{6} = 88$ kg-m^2/s

Final Answer : (2)

20) **Explaining :** Find kinetic energy.
Concept : This question is based on KE in rolling motion.
Solution : When a body rolls over a smooth surface, it has linear K.E. and rotational K.E.

$\therefore E = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$

Where $\omega = \frac{v}{r}$ and $I = \frac{2}{5}mr^2$ for solid sphere

$\therefore K.E. = \frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{2}{5}mr^2\right).\frac{v^2}{r^2}$

$= \frac{1}{2}mv^2 + \frac{1}{5}mv^2 = \frac{7}{10}mv^2 = \frac{7}{10} \times 1 \times 1^2$

$= \frac{7}{10}J$

Final Answer : (3)

21) **Explaining :** Find acceleration of block B
Concept : This question is based on $a_T = \alpha R$
Solution :
Given $a_A = r\alpha = 2\alpha$

$\Rightarrow \alpha = \frac{5}{2}$ rad/s^2

$\Rightarrow a_B = 1 \,.\, \alpha = \frac{5}{2}$ m/s^2

Final Answer : (4)

22) **Explaining :** Find angle covered by grindstone in given interval
Concept : This question is based on If α = constant, apply

$\theta = \left(\frac{\omega_i + \omega_f}{2}\right)t$

Solution :
The angular displacement will be

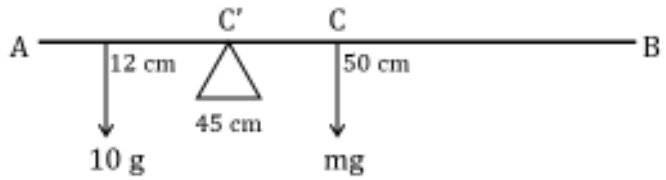
$\Delta\theta = \omega_{avg}\Delta t = \left(\frac{\omega_f + \omega_i}{2}\right)\Delta t$

$= \left(\frac{12.00\,\text{rad/s} + 4.00\,\text{rad/s}}{2}\right)(4.00\,\text{s}) = 32.00$ rad

Final Answer : (4)

23) **Explaining :** Find mass of meter stick
Concept : This question is based on Based on rotational equilibrium
Solution :
Let m be the mass of the metre stick concentrated at C the 50 cm mark as shown in the figure.

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In equilibrium taking moments of forces about C' ; we get
$10g(45 - 12) = mg(50 - 45)$
$10g \times 33 = mg \times 5$
$m = \frac{10 \times 33}{5} = 66g$
Final Answer : (2)

24) **Explaining :** Find angular momentum about (1, 1) cm
Concept : This question is based on $\vec{L} = \vec{r} \times \vec{p}$ (angular momentum).
Solution : $\vec{r} = 9\hat{i} + 5\hat{j}cm$
$\vec{v} = 5\hat{i}cm/s$
$\vec{L} = m(\vec{r} \times \vec{v})$
$= 25 \times 10^{-5}\left(-\hat{k}\right) kgm^2s^{-1}$
Final Answer : (1)

25) **Explaining :** Find K.E. of dics
Concept : This question is based on $K.E_{Rolling} = \frac{1}{2}mv^2\left(1 + \frac{K^2}{R^2}\right)$
Solution :
$KE_{Loop} = \frac{1}{2}mv^2\left(1 + \frac{K^2}{R^2}\right) = 8$
$= \frac{1}{2}mv^2(1 + 1) = 8$
$= \frac{1}{2}mv^2 \times 2 = 8$
$K.E_{loop} = mv^2 = 8\ J$
KE of disc $= \frac{3}{4}mv^2 = \frac{3}{4}(8) = 6\ J$
Final Answer : (3)

26) **Explaining :** Find maximum speed of end P
Concept : This question is based on conservation of mechanical energy
Solution :
At the time of maximum speed of point P, the rod should be vertical
$\Delta K + \Delta U = 0$
$\left[\frac{1}{2}\left(\frac{m\ell^2}{3}\right)\omega^2 - 0\right] + \left[-mg\frac{\ell}{2}\right] = 0$
$\Rightarrow \omega = \sqrt{\frac{3g}{\ell}}$
Velocity of point P, $V_P = \omega\ell = \sqrt{3g\ell}$

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Final Answer : (2)

27) **Explaining :** Select correct option
Concept : This question is based on concepts of pure rolling motion
Solution :
On inclined plane friction is responsible for providing the torque due to which body will rotate.
Final Answer : (2)

28)

Explain Question: The question asks to find torque which can stop the wheel's rotation in one minute.
Concept: Application of Torque.
Solution:

$$\omega_0 = \frac{2\pi N}{t} = \frac{2\pi \times 60}{60} = 2\pi \text{ rad/s}$$

$$\omega = \omega_0 + \alpha t$$

Since $\omega = 0 \Rightarrow \alpha = \frac{-\omega_0}{t} = \frac{-2\pi}{60}$ rad/s^2

Hence $\tau = I\alpha = 2\left(\frac{+2\pi}{60}\right) = \frac{\pi}{15}$ N-m

(negative angular acceleration = angular retardation)

$\tau = \frac{\pi}{15}$N–m (Antitorque)

Final Answer: Option (2)

29) **Explaining :** Find moment of inertia of system about diagonal of square
Concept : This question is based on Moment of inertia of continous system
Solution :

$$I = 0 + 0 + m\left(\frac{\ell}{\sqrt{2}}\right)^2 + m\left(\frac{\ell}{\sqrt{2}}\right)^2$$

= m^2
Final Answer : (2)

30) **Question Explanation :**
Find the minimum horizontal force required to topple a uniform triangular prism resting on a rough surface.
Concept:
This question is based on Rotational Equilibrium - Toppling
Formula : $\vec{\tau} = \vec{r} \times \vec{F}$
Solution:
The tendency of rotating will be about the point C. For minimum force, the torque of F about C has to be equal to the torque of mg about C.

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$\Rightarrow \quad F\left(a\frac{\sqrt{3}}{2}\right) = mg\left(\frac{a}{2}\right)$

$\Rightarrow \quad F = \frac{mg}{\sqrt{3}}$

Final Answer: option (1)

31) **Explaining :** Find MOI of uniform circular disc about given axis.
Concept : This question is based on parallel axis theorem.
Solution :
The moment of inertia about an axis passing through centre of mass of disc and perpendicular to its plane is

$$I_{CM} = \frac{1}{2}MR^2$$

Where M is the mass of disc and R its radius. According to theorem of parallel axis, MI of circular disc about an axis touching the disc at its diameter and normal to the disc is

$$I = I_{CM} + MR^2 = \frac{1}{2}MR^2 + MR^2 = \frac{3}{2}MR^2$$

Final Answer : (3)

32) **Explaining :** Match the column-I with column-II.
Concept : This question is based on kinematics in rotational motion.
Solution :

Angular velocity $\omega = \frac{d\theta}{dt} = 4t + 3 \Rightarrow (\omega)_{t=2} = 4(2) + 3 = 11$ units

Angular acceleration $\alpha = \frac{d\omega}{dt} = 4 \Rightarrow (\alpha)_{t=3} = 4$ unit

At $t = 0, \ \theta = 2(0) + 3(0) + 4 = 4$

At $t = 2, \ \theta = 2(2)^2 + 3(2) + 4$

$= 8 + 6 + 4 = 14$

$$\langle\omega\rangle = \frac{18-4}{2} = \frac{14}{2} = 7 \text{ units}$$

At $t = 0, \ \omega = 4(0) + 3 = 3$ units

At $t = 2, \ \omega = 4(2) + 3 = 11$ units

$$\langle\alpha\rangle = \frac{11-3}{2} = \frac{8}{2} = 4 \text{ units}$$

Final Answer : (1)

33) **Explaining :** To find current in inner and outer wire
Concept : This question is based on Induced parameter
Solution : Current in outer :

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$i = \frac{e}{R} = \frac{1}{R}\frac{d}{dt}$ (B.A)

$i_1 = \frac{1}{50 \times 10^{-3} \times 2\pi b} \times \pi b^2 \frac{dB}{dt}$

$i_1 = \frac{1}{10^{-1}} b \times 0.1$

$i_1 = b = 20 \times 10^{-2} = 2 \times 10^{-1}$ A (clockwise)

current in inner

$i_2 = \frac{1}{R}\frac{d}{dt}$ (B.A')

$i_2 = \frac{1}{50 \times 10^{-3} \times 2\pi a} \times \pi a^2 \frac{dB}{dt}$

$i_2 = \frac{1}{10^{-1}} \times a \times 0.1$

$i_2 = a = 10 \times 10^{-2} = 10^{-1}$ Amp Clockwise

Final Answer : (1)

34) **Explaining :** An aluminium ring B is placed facing an electromagnet A. The current I flowing through A can be increased or decreased.

The question asks: *What happens to the force between A and B when the current is changed?*

Concept : This question is based on

Faraday's & Lenz's Laws

• When the current in A changes, the magnetic flux through the ring B also changes.

• Faraday's Law states that a changing magnetic flux induces an electromotive force (EMF) in the ring.

• Lenz's Law dictates that the induced current's magnetic field opposes the flux change.

• This leads to induced magnetic poles in B that either repel or attract A, depending on whether the current increases or decreases**.**

Solution :

1. Current increases in A ⇒ magnetic flux through B increases.
2. Ring B induces current that generates a magnetic field opposing the increase—that means it forms the same pole as A at the facing side to repel it.
3. Result: A repulsive force pushes the ring away from the electromagnet when current rises.

Final Answer : (2)

35) **Explaining :** Given $\phi = t^3 + 3t - 7$, flux linked with the circuit

Ask : Pattern to graph between time (x-axis) and induced emf.

Concept : This question is based on

(Induced emf) $e = -\frac{d\phi}{dt}$

$y = ax^2 + bx + c$, general equation of parabola.

If $c \neq 0$, then parabola not pass through origin.

Solution : $e = 3t^2 + 3$

It is not in the form $y = 4ax^2$.

Final Answer : (4)

36) **Explanation :** To find the self inductance of the coil

Concept : This question is based on self inductance

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$N\phi = LI$

$\varepsilon = -\frac{Nd\phi}{dt} = \frac{-LdI}{dt}$

Solution : $N\phi = Li \Rightarrow 100 \times 10^{-5} = L \times 5 \Rightarrow L = 0.2$ mH.

Final Answer : (4)

37) **Explaining :** The magnitude of flux passing through the square given

$\vec{B} = \left(a\hat{i} + b\hat{j} + c\hat{k}\right)$ T

Concept : This question is based on $\phi_B = \vec{B} \cdot \vec{A}$

Solution : $\phi = \vec{B}.\vec{A}$

$= \left(a\hat{i} + b\hat{j} + c\hat{k}\right) \cdot \ell^2\hat{k}$

$\phi = c\ell^2$ weber **Final Answer : (4)**

38) **Explaining :** Given : For a solenoid N = 500 turns, □ = 2m and r = 2 cm.

Ask : Self inductance of solenoid.

Concept : This question is based on

Self inductance of solenoid

$L = \frac{\mu_0 N^2 A}{\ell}$

Solution : $L = \frac{\mu_0 N^2 A}{\ell}$

$\Rightarrow L = \frac{4\pi \times 10^{-7} \times 500 \times 500 \times \pi \times (2 \times 10^{-2})^2}{2}$

$\Rightarrow L = 2 \times 10^{-4}$H **Final Answer : (2)**

39) **Question Explanation:** EMF of a rotating copper disc.

Concept: Motional EMF in a Rotating Disc.

Solution:

When a conducting disc of radius r rotates with angular velocity ω in a uniform magnetic field B (parallel to its axis), the induced EMF(e) between the center and the rim is:

$e = \frac{1}{2}Br^2\omega$

Given values:

- Magnetic field, $B = 0.1$ T
- Radius, $r = 0.1$ m
- Frequency, $f = 10$rev/s, so $\omega = 2\pi f = 20\pi$rad/s

Substituting the values:

$$e = \frac{1}{2} \times 0.1 \times (0.1)^2 \times 20\pi$$

$e = \frac{1}{2} \times 0.1 \times 0.01 \times 20\pi = 0.01\pi$ V

Converting to millivolts (1 V = 1000mV):

$e = 0.01\pi \times 1000 = 10\pi$mV

Final Answer: Option (3)

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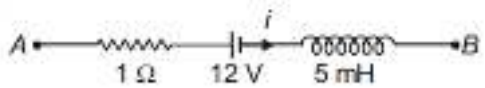
40) **Explaining :** Given :
□ → each side of equivalent triangular loop
R → resistance of loop
$\frac{dB}{dt}$ = 1 T/s varying magnetic field directed into the plane of paper.
Ask : Induced current in the loop.
Concept : This question is based on

$$I = \frac{\varepsilon}{R} = \frac{-d\phi}{Rdt} = \frac{-d\left(\vec{B} \cdot \vec{A}\right)}{Rdt}$$

Area of equilateral triangle = $\frac{\sqrt{3}}{4}\ell^2$.
Solution : $I = \frac{|\varepsilon|}{R} = \frac{AdB}{Rdt} = \frac{\sqrt{3}\ell^2}{4R}$
Final Answer : (1)

41)

Explaining : The network shown in figure is a part of a complete circuit. If at a certain instant, the current i is 4 A and is increasing at a rate of 10^3 A/s. Then $V_B - V_A$ will be



Concept : This question is based on KVL.
Solution : $V_A - 4 \times 1 - 12 - 5 \times 10^{-3} \times 10^3 = V_B$
$-16 - 5 = V_B - V_A$
$V_B - V_A = -21$ volt
Final Answer : (3)

42) **Explaining :** To find the direction of induced current in wires AB and CD if frame is placed in perpendicular magnetic field increasing at a constant rate into the plane of paper.
Concept : This question is based on
Lenz's law states that the induced current. Opposes the change in magnetic flux.
Solution : Due to increasing field, the induced current in both loops individually must be anticlockwise which actually oppose each other. However as the emf induced in larger loop is more, the net current in larger loop is anticlockwise and in smaller loop is clockwise.
Hence, its direction is B to A and D to C.
Final Answer : (4)

43) **Question Explanation:** Current reaches $\frac{1}{4}$ of maximum in 4 s; find time constant.
Concept: In LR growth, current approaches maximum exponentially.
Formula: $I = I_{max}(1 - e^{-t/\tau})$
Calculation:

$$\frac{I}{I_{max}} = \frac{1}{4}$$

$$1 - e^{-t/\tau} = \frac{1}{4}$$

$$e^{-t/\tau} = \frac{3}{4}$$

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$$-\frac{t}{\tau} = \ln\left(\frac{3}{4}\right)$$

$$\tau = \frac{t}{\ln\left(\frac{4}{3}\right)}$$

Given t = 4:

$$\tau = \frac{4}{\ln\left(\frac{4}{3}\right)}$$

Final Answer: Option (3)

44) **Asking about:**
order of current in the given circuits through the battery.

Concept:
LR dc circuit at t = 0 and in steady state

Solution:
(A) at t = 0 inductor behave like open circuit

(i) i_1 and $i_2 = \frac{\varepsilon}{R}$ and $i_3 = \frac{\varepsilon}{2R}$

$\Rightarrow i_2 > i_3 > i_1$, and $i_1 = 0$

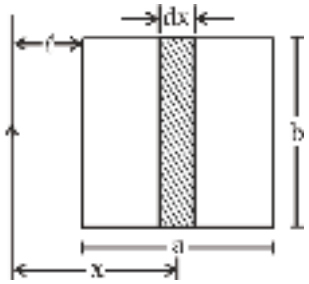
(B) at t = ∞ inductor behaves like short circuit

$$i_1 = \frac{\varepsilon}{2R}, i_2 = \frac{\varepsilon}{R/2} = \frac{2\varepsilon}{R} \quad \text{and } i_3 = \frac{\varepsilon}{R}$$

$\Rightarrow i_2 > i_3 > i_1$

Correct option: (1)

45) **Question Explanation:** Derive M for wire and loop.
Concept: This question is based on Mutual inductance via integration.
Solution:



consider a small strip of the loop with dx at a distance x from the wire. The magnetic field B a distance x is:

$$B = \frac{\mu_0 I}{2\pi x}$$

The small flux $d\phi$ through this strip of area $dA = b \cdot dx$ is:

$$d\phi = B \cdot dA = \frac{\mu_0 Ib}{2\pi x} dx$$

Integrating from x = □ to x = □ + a :

$$\phi = \int_{\ell}^{\ell+a} \frac{\mu_0 Ib}{2\pi x} dx = \frac{\mu_0 Ib}{2\pi} [\ln(x)]_{\ell}^{\ell+a} = \frac{\mu_0 Ib}{2\pi} \ln\left(\frac{\ell + a}{\ell}\right)$$

Since $\phi = MI$, we get:

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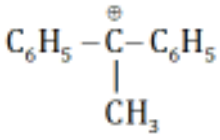
$$M = \frac{\mu_0 b}{2\pi} \ln\left(1 + \frac{a}{\ell}\right)$$

Final Answer: Option (4)

CHEMISTRY

46)

Explaining : Which one is most reactive towards S_N1 reaction?
Concept : This question is based on S_N1 reaction
Solution :
$S_N1 \propto$ stability of intermediate carbocation.

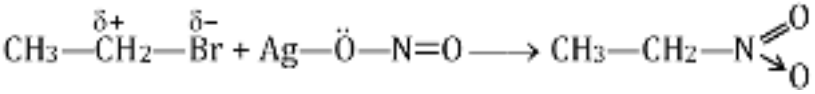
(3)  maximum stable intermediate.
Final Answer : (3)

47) **Explaining :** The above reaction is called -
Concept : This question is based on Swarts reaction
Solution : Formation of alkyl fluoride ⇒ Swarts Reaction
Final Answer : (2)

48) **Explaining :** The reactivity of alkyl halides in E_2 elimination reactions follows the order
Concept : This question is based on E_2 elimination
Solution : R – F < R – Cl < R – Br < R – I
Final Answer : (2)

49) **Explaining :** Which of the following is correct
Concept : This question is based on mixed concepts
Solution : All of these
Final Answer : (4)

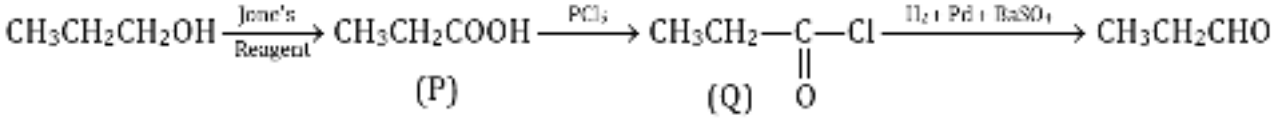
50) **Explaining :** Which of the following reaction is correct :
Concept : This question is based on Alkyl halide reaction
Solution :



Final Answer : (4)

51) **Explaining :** Final product will be
Concept : This question is based on Reaction of alcohol then react with PCl_5.
Solution :

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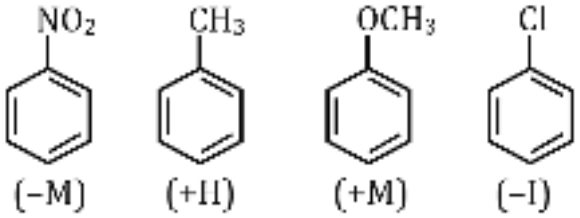


Final Answer : (3)

52) **Explanation :** Question is asking about most reactive towards ESR

Concept : Question is based on $\text{reactivity towards (ESR)} \propto \frac{e^{\ominus} \text{ donating group}}{e^{\ominus} \text{ withdrawing group}}$

Solution :



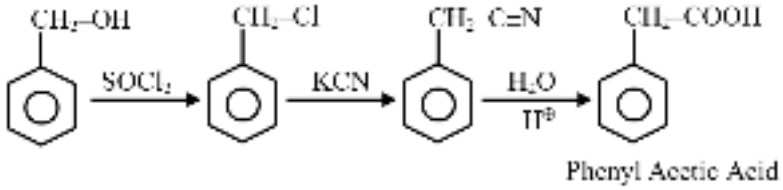
Final Answer : (3)

53) **Explanation :** The question asks to examine four different proposed reaction sequences and determine how many of them correctly result in the formation of benzoic acid.

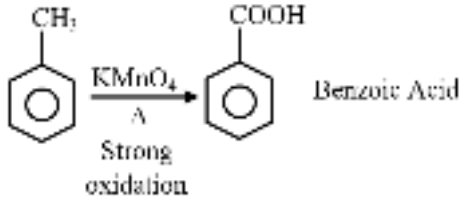
Concept : Alcohol to alkyl halide, Cyanide attack on alkyl halide, Nitride hydrolysis, oxidation alkyl benzene, grignard reagent, reaction with CO_2, F.R. halogenation of alkylbenzene, hydrolysis of $C_6H_5CX_5$.

Solution :

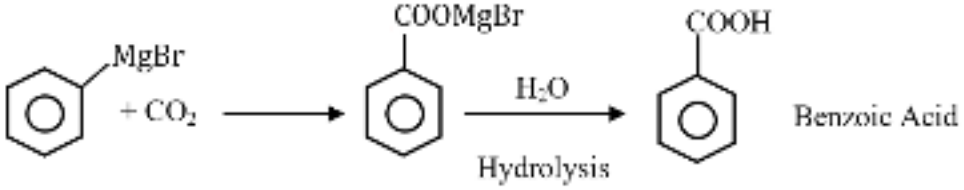
(a) Kushal's Method :



(b) Ayushi's Method :

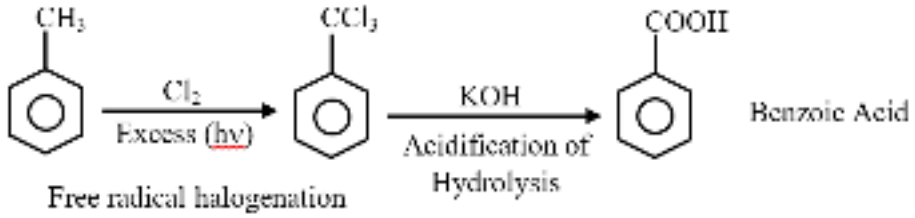


(c) Jagan Jyoti Method :



(d) Anupriya's Method :

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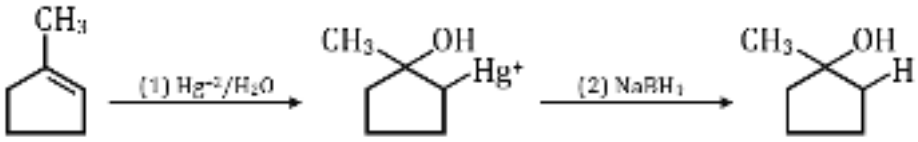
Methods (b), (c) and (d) are correct procedures to synthesize Benzoic Acid.
Final Answer : (3)

54) **Explaining :** In product –OH should be on 3°C.
Concept : This question is based on All are related to EAR.
Solution :

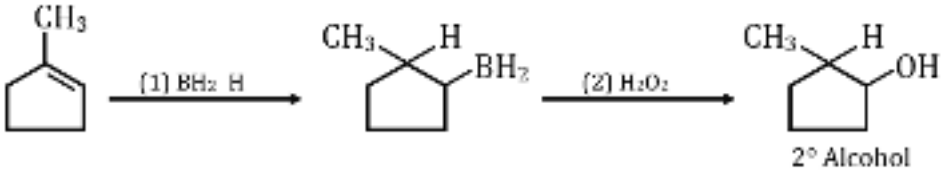
(1)



(2)



(3)



Final Answer : (4)

55) **Explaining :** Which of the following ether can be prepared by Williamson synthesis reaction ?
Concept : This question is based on Williamson synthesis reaction.
Solution : $CH_3 - Cl + Ph - \overset{-}{O}\overset{+}{Na} \rightarrow Ph - O - CH_3$
Final Answer : (3)

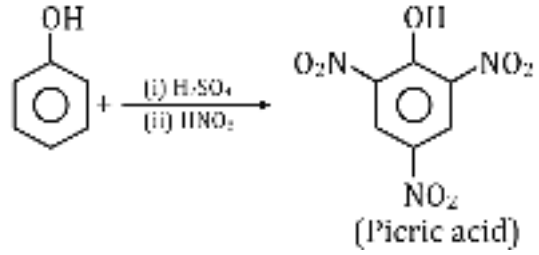
56) **Explaining :** Secondary alcohol on heating with copper at 300°C give.
Concept : This question is based on oxidation reaction.
Solution : $ROH \xrightarrow[573]{Cu} Ketone$
Final Answer : (3)

57) **Explaining :** Most suitable for preparation of ether.
Concept : This question is based on preparation of ether.
Solution :
Williamson ether synthesis
$R\overset{-}{O}\overset{+}{Na} + 1^{o} Alkyl\ Halide \rightarrow R - O - R$

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Final Answer : (2)

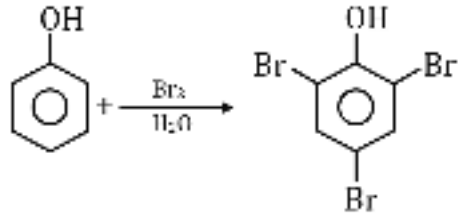
58) **Explaining :** Final product will be.
Concept : This question is based on reaction of phenol.
Solution :



(Picric acid)

Final Answer : (2)

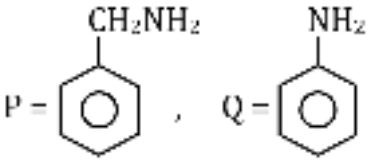
59) **Explaining :** Final product will be
Concept : This question is based on reaction of phenol
Solution :



Final Answer : (4)

60) **Explaining :** Which of the following is R.T. reaction :-
Concept : This question is based on R.T. reaction.
Solution :
(1) → Gatterman
(2) → Reimer-Tiemann
(3) → Kolbe reaction
Final Answer : (2)

61) **Explaining :** P and Q are.
Concept : This question is based on reaction of Amide.
Solution :

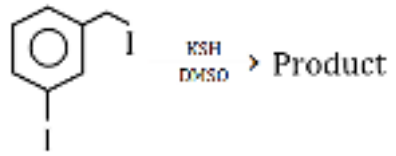


Final Answer : (1)

62) **Explaining :** Correct order of nucleophilicity ?
Concept : This question is based on nucleophilicity.
Solution : $CH_3S^- > CH_3O^- > CH_3\text{–}COO^- > F^-$
Final Answer : (1)

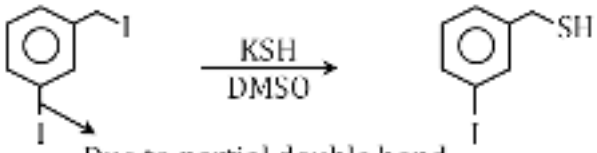
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63) **Explaining :**



Concept : This question is based on S_N2.

Solution :



Due to partial double bond character substitution not possible here in normal

Final Answer : (3)

64)

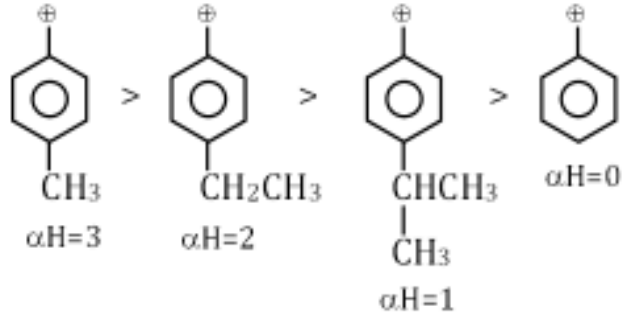
Explaining : Reactivity order for S_N1 reaction :-

Concept : This question is based on S_N1

Solution :

Reactivity of S_N1 ∝ stability of carbocation.

Stability carbocation :

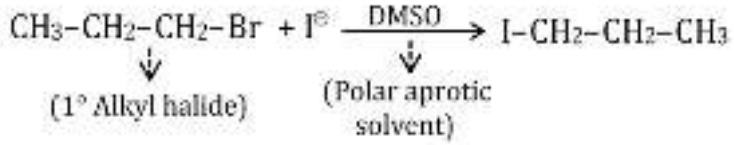


Final Answer : (4)

65) **Explaining :** This reaction would follow which of the following pathway predominantly ?

Concept : This question is based on S_N2.

Solution :



The reaction would follow S_N2 mechanism.

Final Answer : (2)

66) **Explaining :** Which of following is not correctly match

Concept : This question is based on T.L.C.

Solution : Thin layer chromatography – Example of partition chromatography

Final Answer : (3)

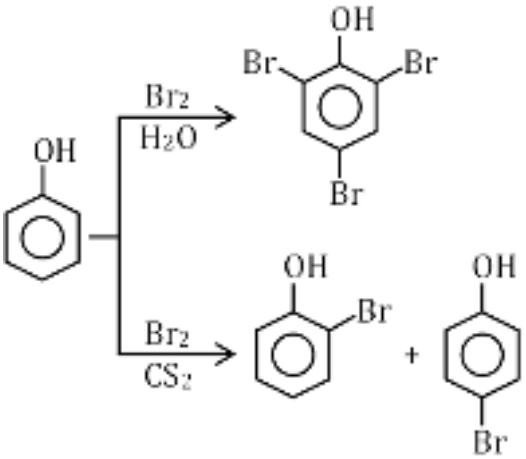
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67) **Explaining :** Match the column-
Concept : This question is based on purification methods.
Solution :
(A) Chloroform & aniline – (P) Distillation
(B) Glycerol & Spentlye – (S) Distillation under reduced pressure
(C) Aniline & Water – (R) Steam distillation
(D) Amino Acid – (Q) Chromatography
Final Answer : (2)

68) **Explaining :** The purification method where solid substance change from vapour state without passing through the liquid state is called as
Concept : This question is based on Sublimation.
Solution : In sublimation the solid substance changes directly to the vapour state.
Final Answer : (1)

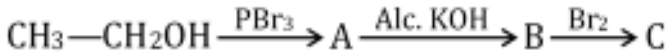
69) **Explaining :** Chloroform (B.P. 334 K) and aniline (B.P. 457 K) are easily separated by the technique of
Concept : This question is based on Distillation.
Solution : Chloroform and aniline has sufficient difference in boiling point and therefore can be separated by distillation.
Final Answer : (2)

70) **Explaining :** What is X and Y in above reaction respectively :-
Concept : This question is based on Phenol reactions.
Solution :

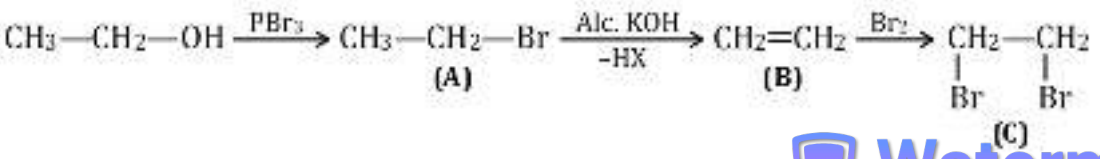


Final Answer : (3)

71) **Explaining :** The compound A, B and C in the reaction sequence



are given by the set :-
Concept : This question is based on Mixed.
Solution :

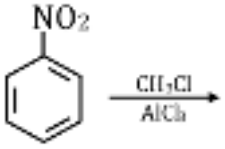


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Final Answer : (3)

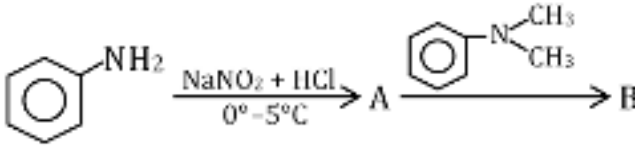
72) **Explaining :** Which of the following will not give isocyanide test :-
Concept : This question is based on Isocyanide test.
Solution : Only 1° Amine gives isocyanide test.
Final Answer : (2)

73) **Explaining :**

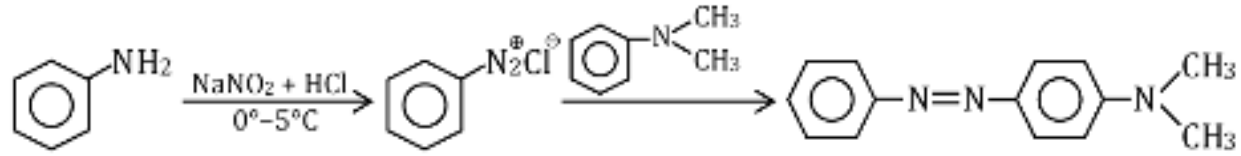


A, A can be :-
Concept : This question is based on ESR.
Solution : Nitrobenzene (-M group) does not give Friedel Craft reaction.
Final Answer : (4)

74) **Explaining :** A coloured product is obtained in the following reaction of Aniline :-



structure of 'B' is :-
Concept : This question is based on Mixed.
Solution :



Final Answer : (2)

75) **Explaining :** Lactose has :-
Concept : This question is based on Lactose.
Solution : β Glycosidic linkage between C_1 of β D galactose and C_4 of β D glucose
Final Answer : (3)

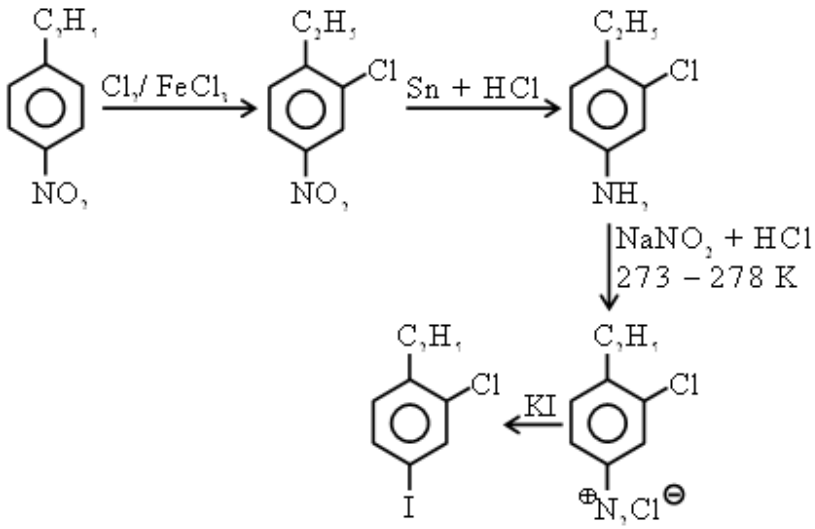
76) **Explaining :** The purest form of vitamin 'C' is -
Concept : This question is based on Vitamin 'C'.
Solution : Ascorbic acid
Final Answer : (3)

77) **Explaining :** Match the column :
Concept : This question is based on Vitamins.
Solution :
(a) Vit A → (r) Xeropthalmia
(b) Vit B12 → (q) Prenicious anaemia
(c) Vit C → (s) Scurvy
(d) Vit D → (p) Osteomalacia

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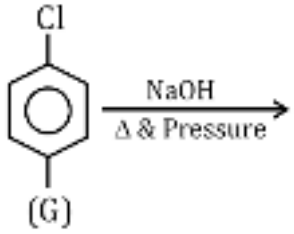
Final Answer : (4)

78) **Explaining :** The major product X formed in the following reaction sequence is :
Concept : This question is based on Mixed concepts.
Solution :



Final Answer : (3)

79) **Explaining :** Rate of reaction is maximum if G is :-
Concept : This question is based on SnAr Mechanism
Solution :



Here ; G must be electron withdrawing group i.e. $-NO_2$
Final Answer : (3)

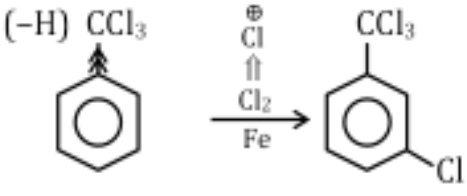
80) **Question Explanation:** Identifying the major product X formed when benzotrichloride ($C_6H_5CCl_3$) undergoes electrophilic chlorination in the presence of an Fe catalyst.

Concept: Reverse hyperconjugation, I effect, meta-directing groups, electrophilic aromatic substitution.

Solution:

A. The $-CCl_3$ group is strongly deactivating due to the –I effect and reverse hyperconjugation.
B. Electron density is withdrawn from the ortho and para positions via resonance, making the meta position relatively more nucleophilic.
C. The catalyst F'e reacts with Cl_2 to generate the CI^+ electrophile, which attacks the meta position.

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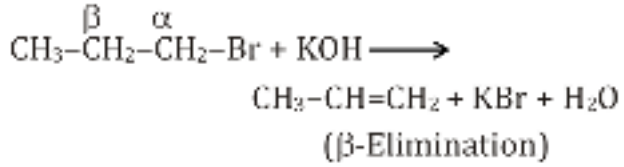


Final Answer : Option (1)

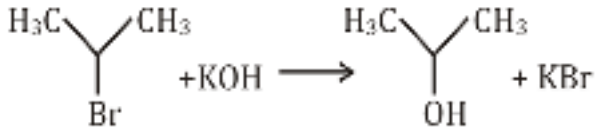
81)

Question Explanation :
The question asks to predict the major organic product formed when 1-chloropropane reacts with methylmagnesium bromide (CH_3MgBr).
Concept :
Reaction of Grignard Reagent with Alkyl Halides, Nucleophilic Substitution (S_{N^2}), Alkane Synthesis
Solution :
The carbanion moiety (CH_3^-) of the Grignard reagent acts as a strong nucleophile and attacks the primary carbon of 1 -chloropropane, displacing the chloride ion via an S_{N^2} mechanism:
$CH_3 - CH_2 - CH_2 - Cl + CH_3MgBr \rightarrow CH_3 - CH_2 - CH_2 - CH_3 + MgBrCl$
This coupling reaction elongates the carbon chain to form n-butane ($CH_3 - CH_2 - CH_2 - CH_3$).
Final Answer : Option (1)

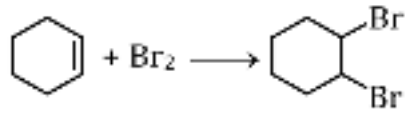
82) **Explaining :** Which is correct statement
Concept : This question is based on type of reaction
Solution :



(β-Elimination)

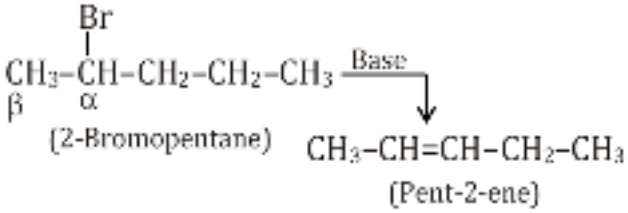


(Nucleophilic substitution reaction)



(Electrophilic addition reaction)
Final Answer : (2)

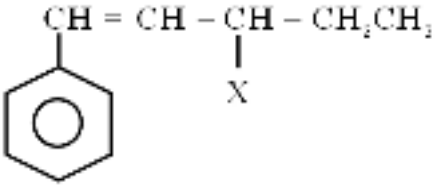
83) **Explaining :** Correct statement for reaction
Concept : This question is based on elimination reaction
Solution :



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Final Answer : (2)

84) **A. Question explanation:-** The given compound is an example of _____.
B. Given Data: We are given the following structure:

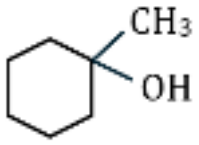


C. Concept: The type of halide is determined by the position of the halogen atom (X) relative to the functional groups or structural features in the molecule.

A. **Aryl halide:** The halogen is directly bonded to an aromatic ring.
B. **Allylic halide:** The halogen is bonded to a carbon atom adjacent to a carbon-carbon double bond.
C. **Vinylic halide:** The halogen is directly bonded to a carbon atom in a carbon-carbon double bond.
D. **Benzylic halide:** The halogen is bonded to a carbon atom adjacent to an aromatic ring.

D. Answer Explanation: In the given structure, the halogen (X) is bonded to a carbon atom that is adjacent to a carbon-carbon double bond. This arrangement fits the definition of an **allylic halide.**
E. Final Answer: Option 2

85) **Question is Asking About:** Most reactive compound (alcohol) with HX.
Concept: Order of rate of reaction of Alcohol with HX is 3° > 2° > 1°.
Solution/Explanation/Calculation:

Since 

is 3° Alcohol hence is most reactive with HX.
⇒ Phenol do not react with HX.
Hence, option (1) is correct.
Final Answer : (1)

86)

Question Explanation: Identify the name of the reaction in which **sodium phenoxide reacts with CO_2** to form **sodium salicylate**.

Concept: Kolbe–Schmitt Reaction, Carboxylation of Phenol

Solution: Sodium phenoxide reacts with **CO_2 at about 140°C under pressure** to form **sodium salicylate**, which on acidification gives **salicylic acid**. This carboxylation reaction is known as the **Kolbe–Schmitt reaction**.

Final Answer: Option (2)

87)

Question Explanation: The question asks for the identification of products A and B formed by

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the dehydration of ethanol (CH_3CH_2OH) using concentrated H_2SO_4 at two different temperatures: 443K and 413K.

Concept: Acid-catalyzed dehydration, Temperature-dependence, Elimination (E1/E2), Nucleophilic substitution (S_N2), Ethene formation, Ether synthesis.

Solution:
The reaction outcome depends strictly on the temperature:

A. **At 443K (High Temp):** Intra-molecular dehydration occurs. A water molecule is eliminated from a single ethanol molecule to form an alkene.

$$CH_3CH_2OH \xrightarrow[443K]{H_2SO_4} CH_2 = CH_2 + H_2O$$

Product A: Ethene ($CH_2 = CH_2$)

A. **At 413K (Low Temp):** Inter-molecular dehydration occurs. Two ethanol molecules combine to eliminate one water molecule, forming an ether.

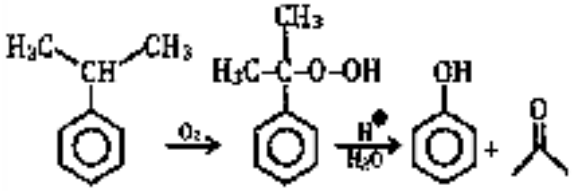
$$2CH_3CH_2OH \xrightarrow[413K]{H_2SO_4} CH_3CH_2 - O - CH_2CH_3 + H_2O$$

Product B: Ethoxyethane (Diethyl ether)

Final Answer: Option (4)

88)

Explain Question: The structure of intermediate A in the given reaction is.
Concept: Cumene hydroperoxide method.
Solution:



Final Answer: option (2)

89) **Explaining :** Final product will be
Concept : This question is based on Ether cleavage
Solution :

$$CH_3–CH_2–CH_2–O–C(CH_3)_3 \xrightarrow[\Delta]{\text{excess HI}} CH_3–CH_2–CH_2–OH + (CH_3)_3C–I$$

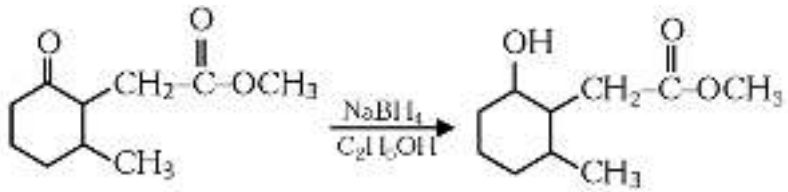
Δ | HI

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CH_3–CH_2–CH_2–I

Final Answer : (1)

90) **Explain question:** Find out the product formed in the following chemical reaction?
Concept: Reduction with $NaBH_4$
Solution:



$NaBH_4$ reduces aldehyde/ketone but does not reduce ester.
Final Answer. Option (4)

BIOLOGY

91) **Solution :** Statements 1, 2, 4 and 5 are correct. Uracil occurs in RNA, whereas thymine occurs in DNA.
Final Answer : (1)

92) **Solution :** The 2′-OH contributes to RNA's reactive and unstable nature, whereas DNA lacks this group.
Final Answer : (3)

93) **Solution :** Primary transcripts contain both exons and introns. Splicing removes introns and joins exons.
Final Answer : (2)

94) **Solution :** The i, z, y and a genes respectively code for repressor, β-galactosidase, permease and transacetylase.
Final Answer : (1)

95) **Solution :** Capping occurs at the 5′ end; tailing occurs at the 3′ end. Splicing removes introns and joins exons.
Final Answer : (1)

96) **Solution :** Heterochromatin is densely packed and transcriptionally inactive, whereas euchromatin is transcriptionally active.
Final Answer : (2)

97) **Solution :** Translation begins at AUG, followed by sequential tRNA binding and peptide-bond formation; termination occurs when release factor binds the stop codon.
Final Answer : (2)

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98) **Solution :** Total DNA was fragmented, cloned and amplified, sequenced, arranged using overlaps, aligned using computer programs and subsequently annotated.
Final Answer : (1)

99) **Solution :** Eukaryotic primary transcripts contain exons and introns; splicing removes introns and joins exons to generate functional RNA.
Final Answer : (2)

100) **Solution :** Complementarity gives TACGTACG, and because the two DNA strands are antiparallel, it is written 5′→3′ as **5′-TACGTACG-3′**.
Final Answer : (1)

101) **Solution :** Lactose/allolactose interacts with the repressor, inactivates it and permits RNA polymerase access to the promoter.
Final Answer : (2)

102) **Solution :** Replication and transcription can be conceptualised on the basis of complementarity because one nucleic acid is copied into another. Translation transfers information from nucleotide polymer to amino-acid polymer, for which nucleotide–amino acid complementarity does not exist.
Final Answer : (1)

103) **Solution :** Change of a single base pair in the β-globin gene resulting in glutamate-to-valine substitution as a classical example of point mutation and sickle-cell anaemia
Final Answer : (2)

104) **Solution :** The template strand is 3′→5′ and directs RNA synthesis; the coding strand is 5′→3′ and resembles RNA except that DNA contains thymine instead of uracil.
Final Answer : (2)

105) **Solution :** Variation in VNTR repeat copy numbers generates fragments of different sizes and consequently a characteristic banding pattern
Final Answer : (2)

106) **Solution :** UUU coding for phenylalanine from bacteria to humans as an example and mentions exceptions in mitochondrial codons and some protozoans
Final Answer : (2)

107) **Solution :** In bacteria, transcription and translation occur in the same compartment and bacterial mRNA does not require processing to become active, allowing the two processes to be coupled.
Final Answer : (3)

108) **Solution :** DNA fingerprinting identifies variation in specific repetitive DNA regions.

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Satellite DNA includes highly polymorphic repetitive sequences such as VNTRs, which form the basis of DNA fingerprinting.
Final Answer : (1)

109) **Solution :** C = G = 20%, therefore A + T = 60%. Since A = T, A = 30%.
Final Answer : (2)

110) **Solution :** $6.6 \times 10^{9} \times 0.34 \times 10^{-9} = 2.244$ m ≈ 2.2 m
Final Answer : (2)

111) **Solution :** AUG codes for methionine, UUU for phenylalanine, UUC for phenylalanine and UAG is a stop codon. Thus, **3 amino acids** are incorporated.
Final Answer : (3)

112) **Solution :** Associates loosely packed, lightly staining euchromatin with transcriptional activity.
Final Answer : (2)

113) **Solution :** Translation can begin even before transcription of the complete mRNA is finished.
Final Answer : (1)

114) **Solution :** The **3′→5′ template strand** is used for RNA synthesis. The coding strand is not the template.
Final Answer : (4)

115) **Solution :** DNase digestion inhibited transformation, indicating that DNA was responsible for transformation. Proteases and RNases did not affect transformation.
Final Answer : (3)

116) **Solution :** DNA contains phosphorus, whereas protein does not. Therefore, radioactive phosphorus was used to label DNA. Radioactive sulfur was used to label protein because protein contains sulfur while DNA does not.
Final Answer : (1)

117) **Solution :** : Prokaryotes do not have a defined nucleus. Their DNA is organised in a region called the nucleoid.
Final Answer : (2)

118) **Solution :** Lysine and arginine
Final Answer : (2)

119) **Solution :** The two DNA strands have opposite polarity. If one strand runs 5′→3′, the

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other runs 3′→5′.
Final Answer : (2)

120) **Solution :** S strain bacteria possess a mucous polysaccharide coat and are virulent, whereas R strain bacteria lack this coat and are non-virulent.
Final Answer : (2)

121) **Solution :** Negatively charged DNA wraps around a positively charged histone octamer to form a nucleosome.
Final Answer : (2)

122) **Solution :** A typical nucleosome contains approximately 200 bp of DNA helix
Final Answer : (4)

123) **Solution :** Chromosome 1 as having most genes, Y as having the fewest, and identifies SNPs as single nucleotide polymorphisms.
Final Answer : (1)

124) **Solution :** The purines are adenine and guanine. Cytosine, thymine and uracil are pyrimidines.
Final Answer : (3)

125) **Solution :** There is a single DNA dependent RNA polymerase that catalyses transcription of all types of DNA in bacteria.
Final Answer : (2)

126) **Solution :** The three critical research areas are:

A. Best catalyst → Improved organism or pure enzyme.
B. Optimal conditions → Engineering suitable conditions for catalyst activity.
C. Downstream processing → Purification of the desired protein or organic compound

Final Answer : (1)

127) **Solution :** Sucrose provides a carbon source and energy for cultured plant tissues. The medium also requires mineral salts, vitamins, amino acids and growth regulators.
Final Answer : (2)

128) **Solution :** Both statements are correct. However, the reason only describes the method; the genetic identity is due to clonal propagation, not merely the fact that tissue culture is used.
Final Answer : (2)

129) **Solution :** Pomato was obtained by **fusion of tomato and potato protoplasts**, followed by growth of the hybrid cells into a plant.

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Final Answer : (2)

130) **Solution :** The sequence is:
Alkaline gut → crystal solubilisation → protoxin activation → binding to midgut cells → pore formation → swelling/lysis → insect death.
Final Answer : (1)

131) **Solution :**

A. *cryIAc* + *cryIIAb* → cotton bollworms
B. *cryIAb* → corn borer

Final Answer : (2)

132) **Solution :** RNAi is the strategy used to protect tobacco from the nematode ***Meloidogyne incognita***.
Final Answer : (2)

133) **Solution :** The meristem is free of virus, making meristem culture useful for obtaining healthy plants.
Final Answer : (1)

134) **Solution :** Insulin is synthesised as proinsulin, which contains an extra C-peptide. During maturation, the C-peptide is removed, producing mature insulin.
Final Answer : (2)

135) **Solution :** Gene therapy involves introducing a normal functional gene into a person's cells or tissues to compensate for a non-functional gene.
Final Answer : (2)

136) **Solution :** First clinical gene therapy was given in 1990 to a 4-year-old girl suffering from ADA deficiency.
Final Answer : (3)

137) **Solution :** PCR amplifies specific nucleic acid sequences, making even very low amounts of pathogen DNA detectable.
Final Answer : (1)

138) **Solution :** ELISA is based on antigen–antibody interaction.
Final Answer : (1)

139) **Solution :** Insulin production uses recombinant DNA technology. RNAi is associated with gene silencing, not recombinant insulin production.
Final Answer : (4)

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140) **Solution :** Modifying genes that affect insulin-like growth factor and studying resulting biological effects to understand the normal physiology and developmental role of that factor
Final Answer : (2)

141) **Solution :** A-P, B-Q, C-R, D-S

Column I	Column II
A. Rosie	P. First transgenic cow
B. Alpha-1-antitrypsin	Q. Treatment of emphysema
C. Alpha-lactalbumin	R. Human protein-enriched milk
D. Insulin-like growth factor	S. Study of normal growth

Final Answer : (1)

142) **Solution :** A, B and C are correct. Statement D is not the described role of GEAC.
Final Answer : (2)

143) **Solution :** A foreign gene can be introduced into an animal to produce a useful biological product, such as a human protein
Final Answer : (1)

144) **Solution :** Biopiracy involves use without proper authorisation and without compensatory payment. Statement C is incorrect because proper authorisation is absent in the described definition
Final Answer : (2)

145) **Solution :** A, B and D are correct. Ethical issues also concern living organisms, ecosystems, ownership, authorisation and benefit sharing. Hence C is incorrect.
Final Answer : (2)

146) **Solution :** I. The bacterial cells must be made competent to take up DNA.
II. The cell is treated with specific concentration of a divalent cation such as *Calcium* (Ca^{2+}) to the efficiency with which DNA enters.
III In *Microinjection* method, recombinant DNA is directly injected into the nucleus of an animal cell.
Final Answer : (3)

147) **Solution :** Stanley Cohen and Herbert Boyer in 1972 constructed the first recombinant DNA using a native plasmid of *Salmonella typhimurium*.
Final Answer : (1)

148) **Solution :** Isolation of the plasmid from *Salmonella typhimurium* → Cutting the plasmid and the antibiotic resistance gene using restriction enzymes → Insertion of the antibiotic resistance gene into the plasmid → Introduction of recombinant plasmid into a host bacterium → Selection of transformed cells using antibiotics.

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Final Answer : (1)

149) **Solution :** Restriction endonuclease — Cuts DNA at specific recognition sequences
DNA ligase — Joins DNA fragments
Methylase — Adds methyl groups to DNA
Plasmid — Circular DNA vector used in genetic engineering
Recognition sequence — Specific DNA sequence recognized by restriction enzymes
Final Answer : (1)

150) **Solution :** Both types of transformants retain ampicillin resistance. Further screening on tetracycline is required to distinguish them.
Final Answer : (3)

151) **Solution :**

A. Statement I is true.
B. Statement II is false — the extension step usually occurs at ~72°C, which is optimal for Taq DNA polymerase, not 55°C (that's usually the annealing temperature).

Final Answer : (2)

152) **Solution :**

A. Statement I is correct – Bioreactors provide controlled conditions (pH, temperature, oxygen, nutrients) to maximize cell growth and protein yield.
B. Statement II is incorrect – Small-scale cultures cannot produce large amounts of recombinant proteins; that's why bioreactors are necessary for commercial-scale production.

Final Answer : (2)

153) **Solution :** Clinical trials are essential for any therapeutic product before marketing. Statement II is incorrect.
Final Answer : (3)

154) **Solution :** It is a thermostable DNA polymerase (isolated from a bacterium, *Thermus aquaticus*), which remain active during the high temperature induced denaturation of double stranded DNA.
Final Answer : (3)

155) **Solution :** Insertion within the lacZ gene disrupts its function, so β-galactosidase is not produced — enabling blue-white screening for identifying recombinants.
Final Answer : (3)

156) **Solution :** In gel electrophoresis the DNA fragments separate according to the size (smaller the fragment size, the faster it moves) this is called sieving effect.
Final Answer : (1)

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157) **Solution :** It is preferable for a cloning vector to have a single unique recognition for restriction enzyme because multiple restriction sites can cause the vector to be cut into multiple fragments, complicating cloning.
Final Answer : (2)

158) **Solution :**

A. DNA moves towards the anode (positive electrode) because of its negative charge, so A is wrong.
B. Restriction enzymes cut at specific palindromic sequences, not randomly, so B is wrong.
C. Using the same enzyme on both vector and insert creates compatible sticky ends, allowing ligation (correct)
D. DNA ligase joins DNA fragments by forming phosphodiester bonds; it does not cut DNA, so D is wrong.

Final Answer : (3)

159) **Solution :** For transformation, microparticles coated with DNA to be bombarded with gene gun are made up of gold or tungsten.
Final Answer : (2)

160) **Solution :** T-DNA naturally integrates into plant cells and has been engineered for gene delivery.
Final Answer : (2)

161) **Solution :** Both A and R are true, but R is not the correct explanation of A.
Final Answer : (2)

162) **Solution :** X is amp^R gene and Y represents tet^R gene. These are the selectable markers.
Final Answer : (2)

163) **Solution :** When cold ethanol is added, DNA precipitates out as white threads — this is called spooling
Final Answer : (3)

164) **Solution :** Curd contains numerous lactic acid bacteria (LAB) or *Lactobacillus*. These bacteria produce acids that coagulate and digest milk proteins. It improves its nutritional quality by increasing vitamin B_{12}.
Final Answer : (2)

165) **Solution :** Streptokinase are produced by Streptococcus and are used as a 'clot buster' to remove clots from the blood vessels of patients who have myocardial infarction.
Final Answer : (1)

166) **Solution :** Large holes in 'Swiss cheese' are due to production of CO_2 by *Propionibacterium sharmanii* (a bacterium).

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Final Answer : (1)

167) **Solution :** Monascus purpureus is a yeast used in the production of statins. Statins are used in lowering blood cholestrol.
Final Answer : (4)

168) **Solution :** Microbial biocontrol agent that can be introduced in order to control butterfly caterpillars is the bacteria Bacillus thuringiensis (Bt). They are sprayed onto vulnerable plants, where they are eaten by the insect larvae. In the gut of the larvae, the toxin is released and the larvae get killed. The bacterial disease will kill the caterpillars, but leave other insects unharmed
Final Answer : (3)

169) **Solution :** Biogas is used for various purposes, as fuel for heating, cooking, lighting, power for irrigation and as an alternative to kerosene, firewood dung cakes.
Final Answer : (2)

170) **Solution :** The bacterium involved in lactic acid production is *Lactobacillus*. The fungus involved in Cyclosporin A production is *Trichoderma polysporum*. Penicillin is obtained from Penicillium notatum, a fungus.
Final Answer : (1)

171) **Solution :** The label A represents sludge, label B represents methane and carbon dioxide, and the label C represents dung and water.
Final Answer : (2)

172) **Solution :** Biopesticides pose less threat than conventional pesticides. Only targeted pest and closely related organisms are affected by biopesticides in contrast to synthetic pesticides that affect a large spectrums of animals. Microorganisms used as biopesticide include viruses, bacteria, and protozoa is correct statement but not the correct explanation of assertion.
Final Answer : (2)

173) **Solution :** Symbiotic association is exhibited by Mycorrhiza and Rhizobium. Heterocyst BGA and Yeast are non-symbiotic.
Final Answer : (4)

174) **Solution :** The major component of biogas is methane (about 50-68%) which is highly inflammable . The other gases are carbon dioxide (25 - 35%), hydrogen (1 - 5%), nitrogen (2 - 7%), oxygen (0 - 0.1%) and rarely hydrogen sulfide. Biogas is a “mixture of gases” produced from degradable organic matter by the activity of various anaerobic bacteria that offers a low cost alternative for energy requirements
Final Answer : (4)

175) **Solution :** Primary sludge is used for the preparation of compost, manure and biogas production. Primary sludge is a result of the capture of suspended solids and organics in the

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primary treatment process through gravitational sedimentation. The secondary treatment process uses microorganisms to consume the organic matter in the wastewater. The microorganisms feed on the biodegradable material in the wastewater in the aeration tank then flow into a secondary clarifier where the biomass settles out and removed as secondary sludge.
Final Answer : (4)

176) **Solution :** Antibiotics are medicines that are produced by certain microorganisms to kill other disease causing microorganisms. These medicines are commonly obtained from bacteria and fungi
Final Answer : (3)

177) **Solution :** Yeast is used in the making of alcohol. For many years, the only source of ethanol for industrial use was from the fermentation of sugars by yeast. The basic reaction of this conversion involves the breakdown of glucose into carbon dioxide and ethanol. The glucose is converted to pyruvic acid and the pyruvic acid may be converted into many different end products such as ethanol, lactic acid, etc.
Final Answer : (3)

178)

Solution/Explanation/Calculation:
Eco RI is a restriction enzyme that specifically recognizes and cuts the DNA sequence GAATTC between the G and A bases, resulting in sticky ends.

Conclusion:
Hence, option (2) is correct..

179) **Explanation :**
When DNA is stained with ethidium bromide and exposed to UV radiation, it fluoresces with a bright orange color. Ethidium bromide intercalates between the DNA base pairs and emits orange fluorescence under UV light.

The correct answer is: 3. Bright orange colour

180) **Solution:** *Lactobacillus* → Curd
Saccharomyces cerevisiae → Bread (fermentation)
Aspergillus niger → Citric acid
Acetobacter aceti → Acetic acid
Final Answer : (2)

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